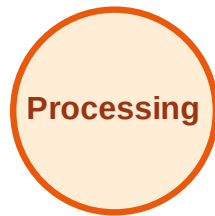


BFS Traversal

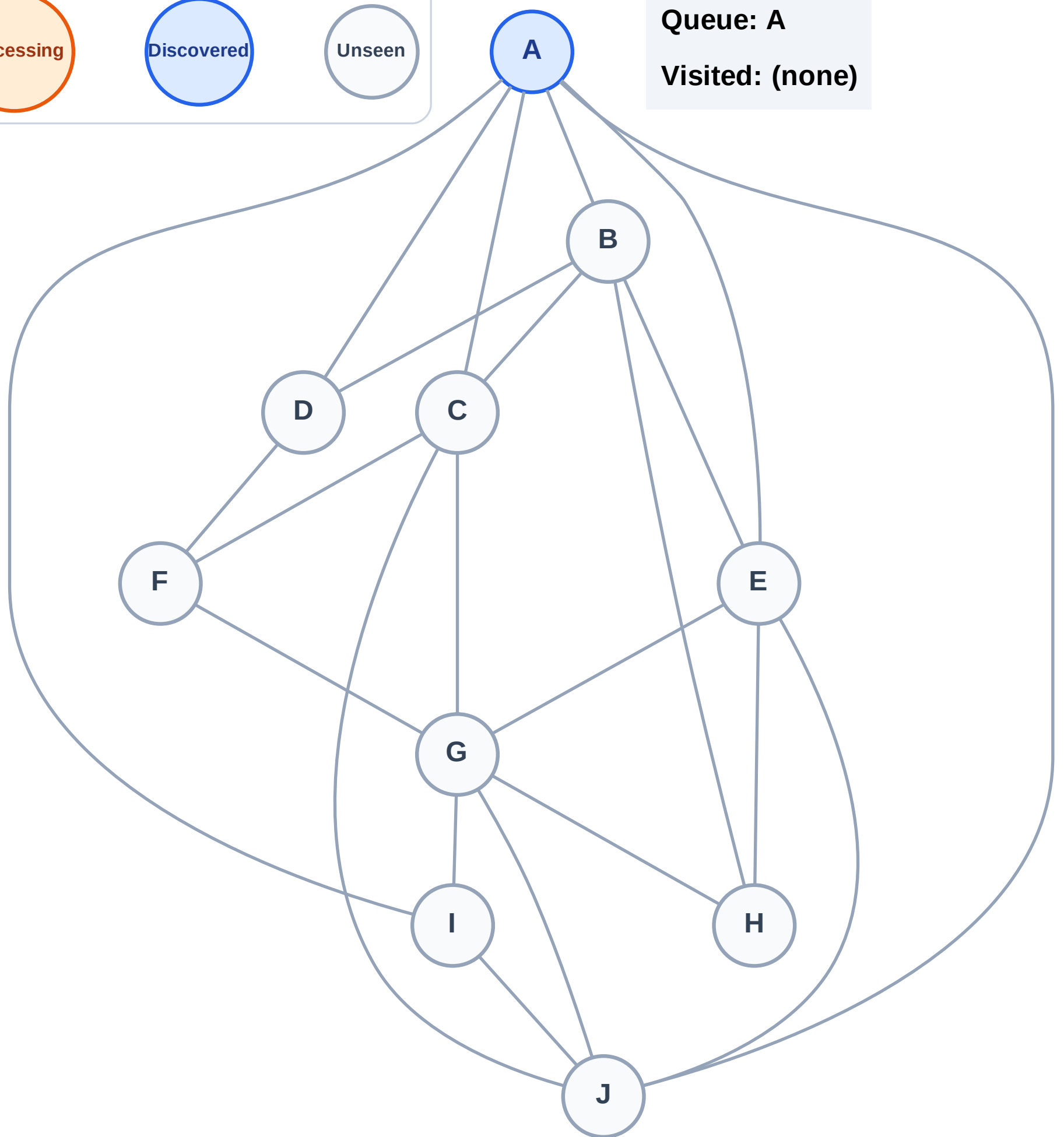
Step 1: Start at A

Legend



Queue: A

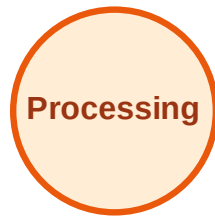
Visited: (none)



BFS Traversal

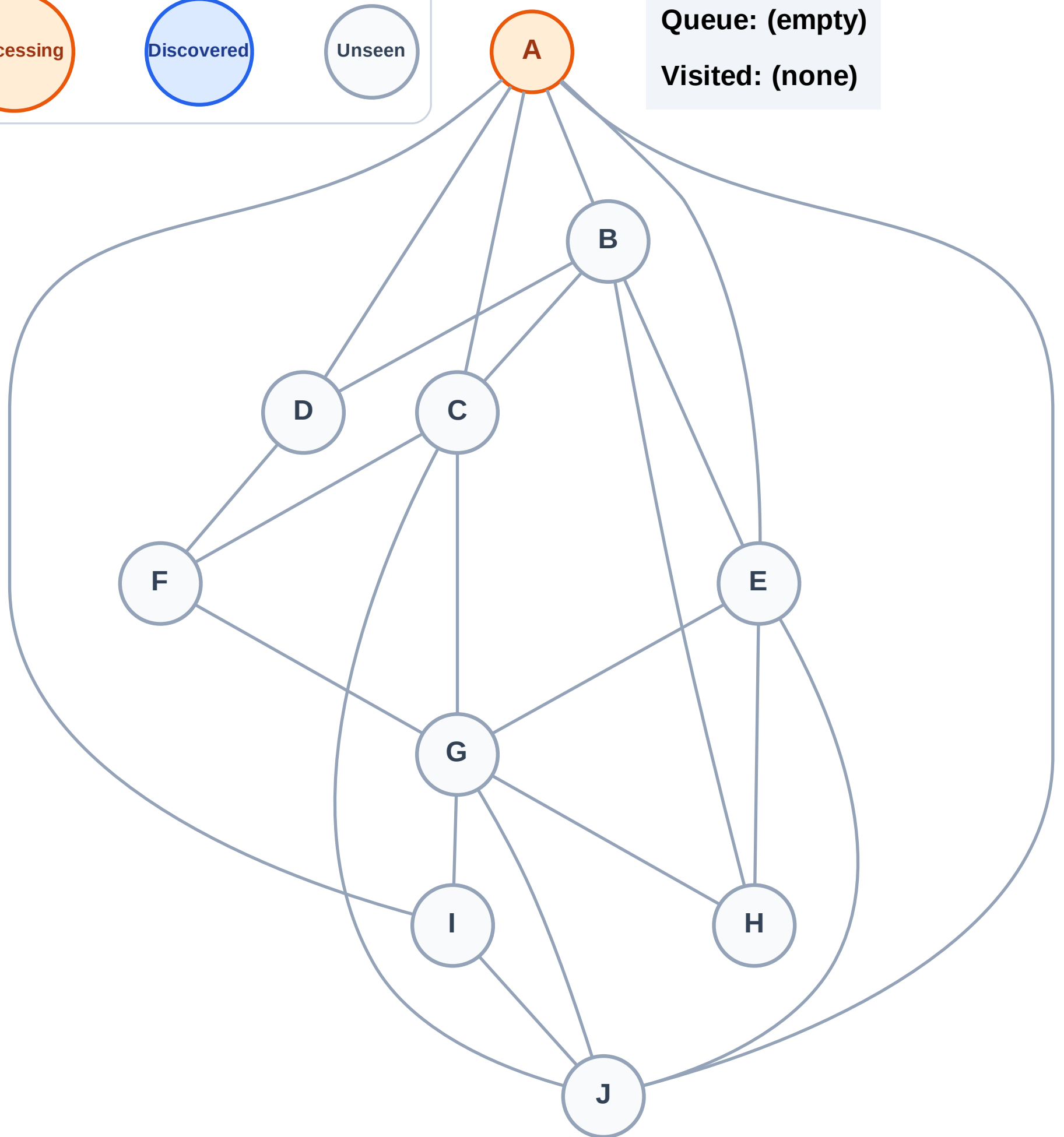
Step 2: Process node A

Legend



Queue: (empty)

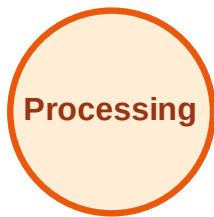
Visited: (none)



BFS Traversal

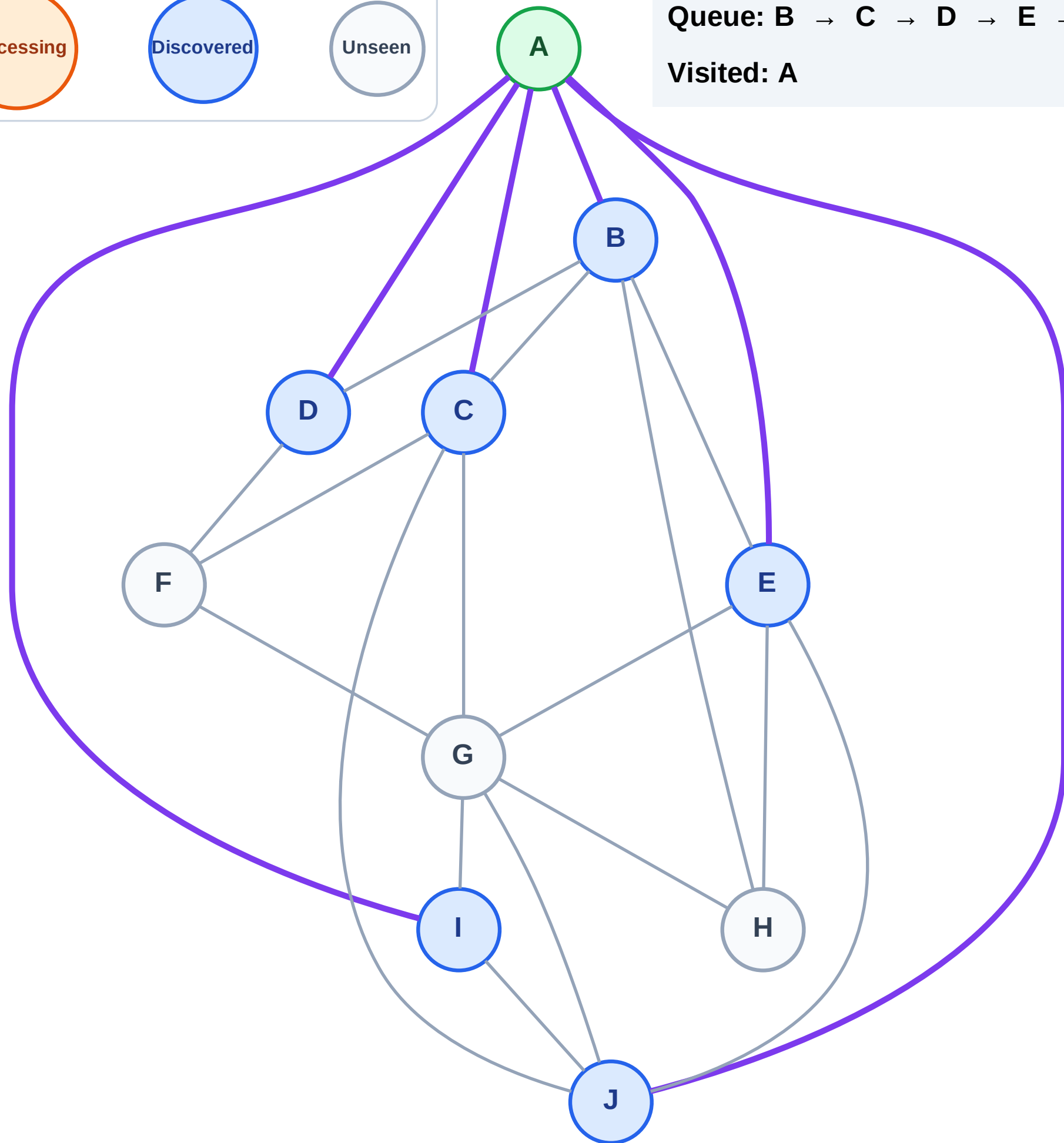
Step 3: Discover B, C, D, E, I, J from A

Legend



Queue: B → C → D → E → I → J

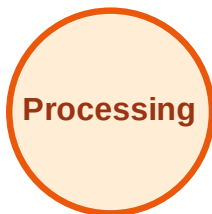
Visited: A



BFS Traversal

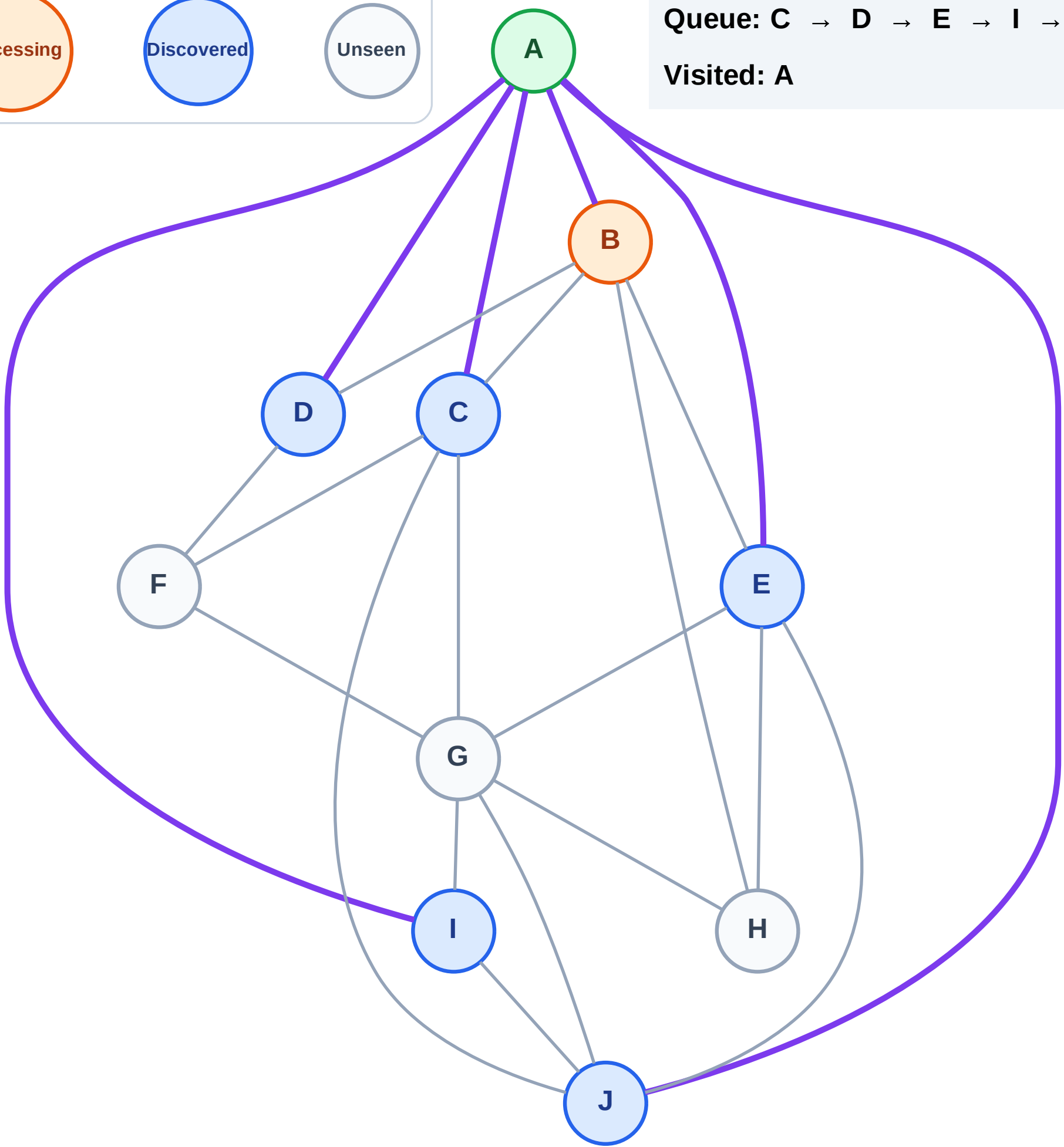
Step 4: Process node B

Legend



Queue: C → D → E → I → J

Visited: A



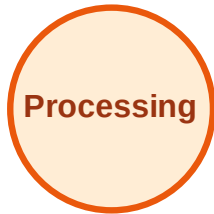
BFS Traversal

Step 5: Discover H from B

Legend



Complete



Processing



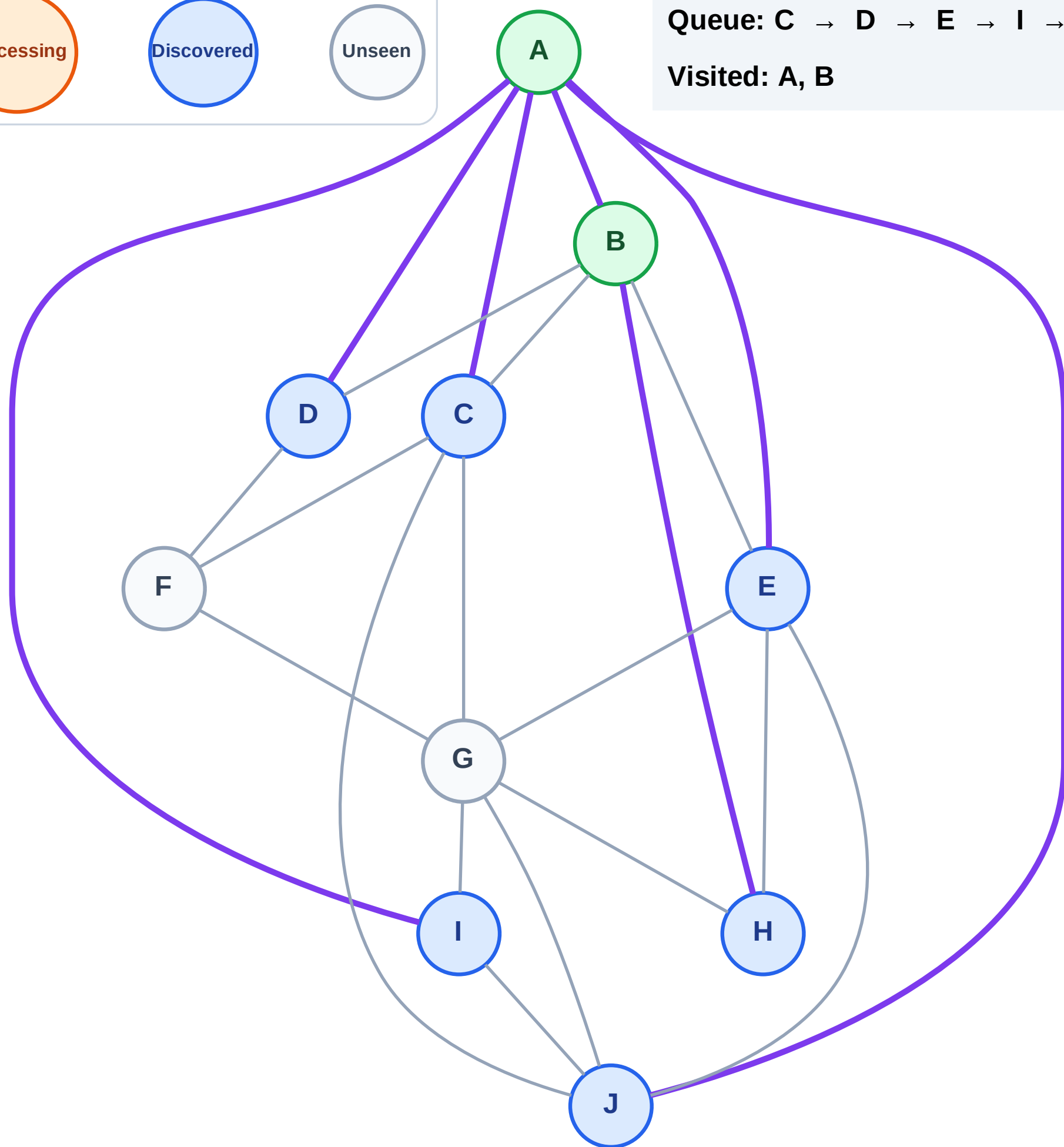
Discovered



Unseen

Queue: C → D → E → I → J → H

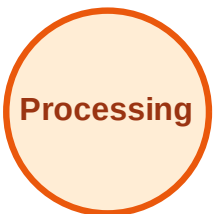
Visited: A, B



BFS Traversal

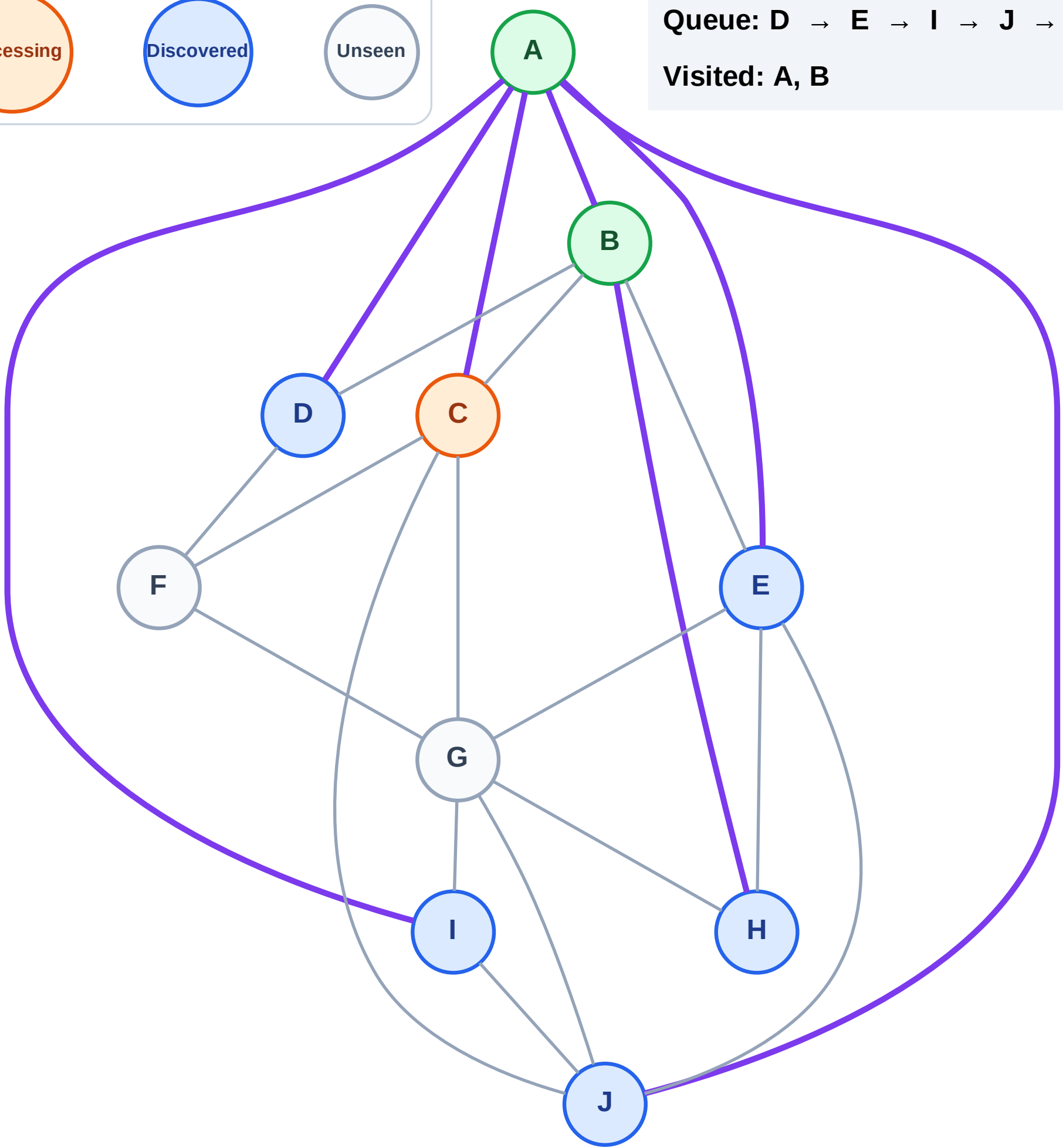
Step 6: Process node C

Legend



Queue: D → E → I → J → H

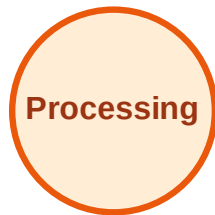
Visited: A, B



BFS Traversal

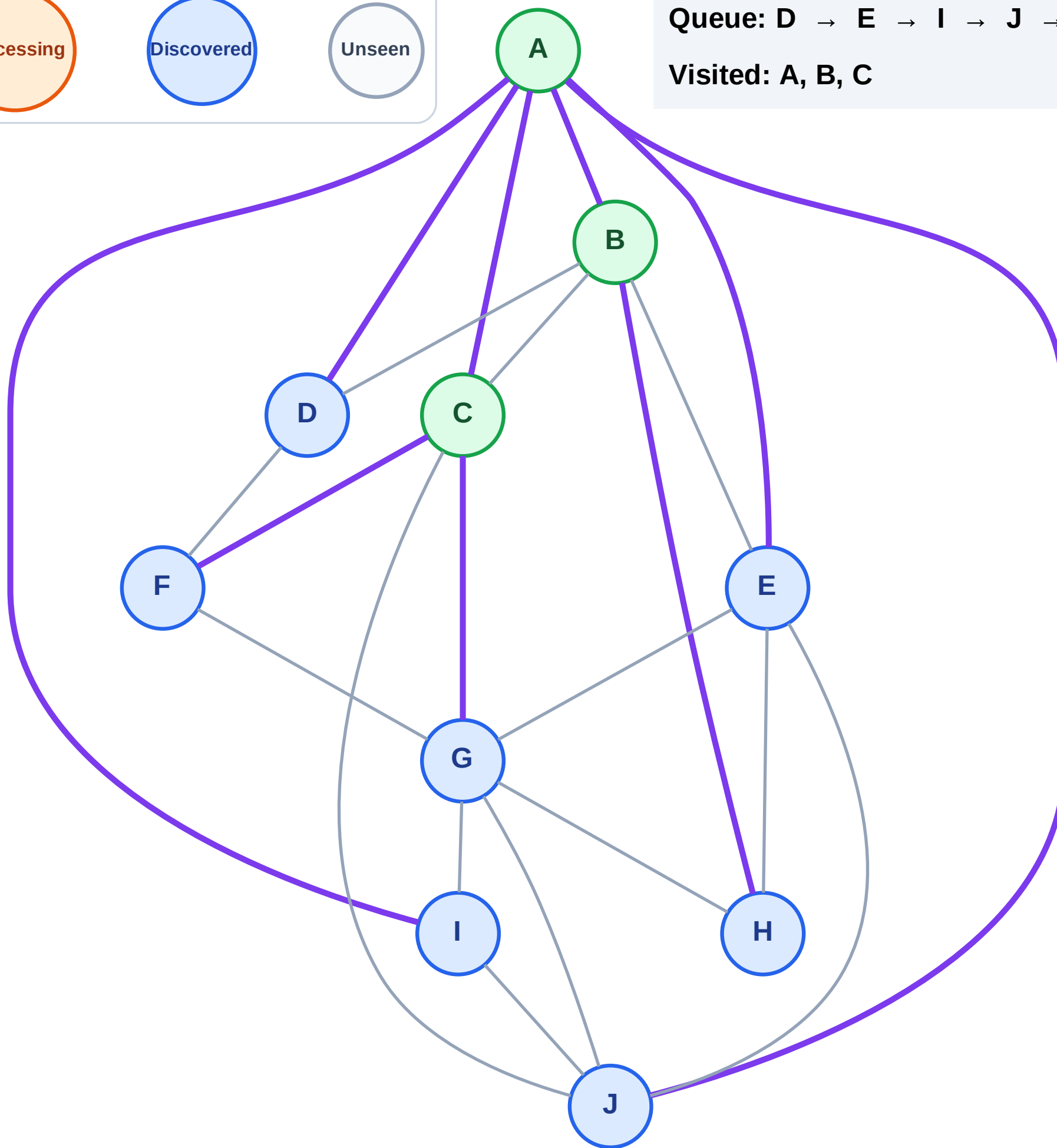
Step 7: Discover F, G from C

Legend



Queue: D → E → I → J → H → F → G

Visited: A, B, C



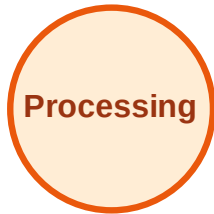
BFS Traversal

Step 8: Process node D

Legend



Complete



Processing



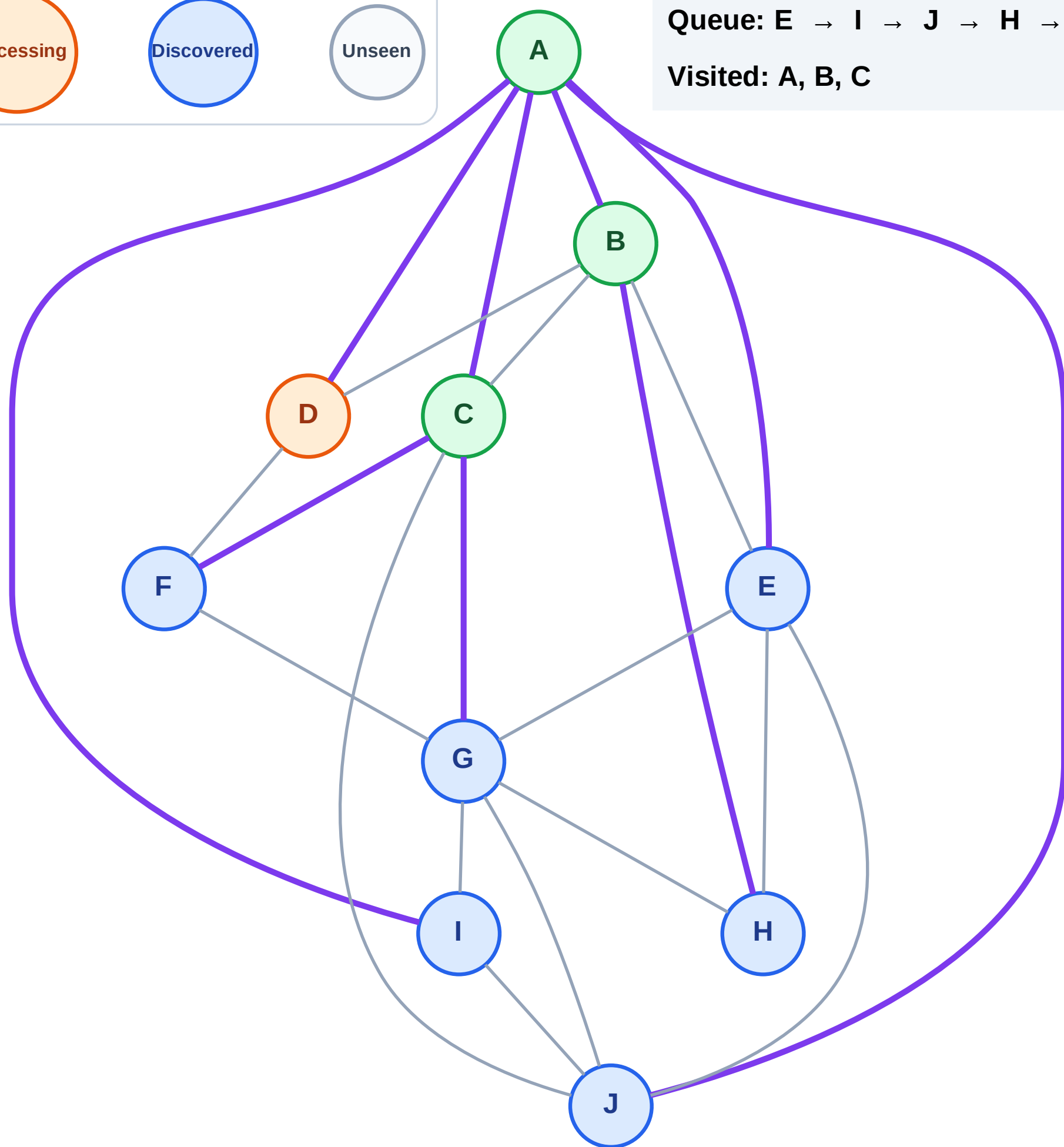
Discovered



Unseen

Queue: E → I → J → H → F → G

Visited: A, B, C



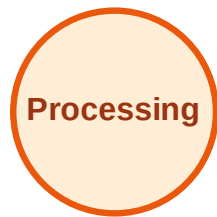
BFS Traversal

Step 9: No new neighbors from D

Legend



Complete



Processing



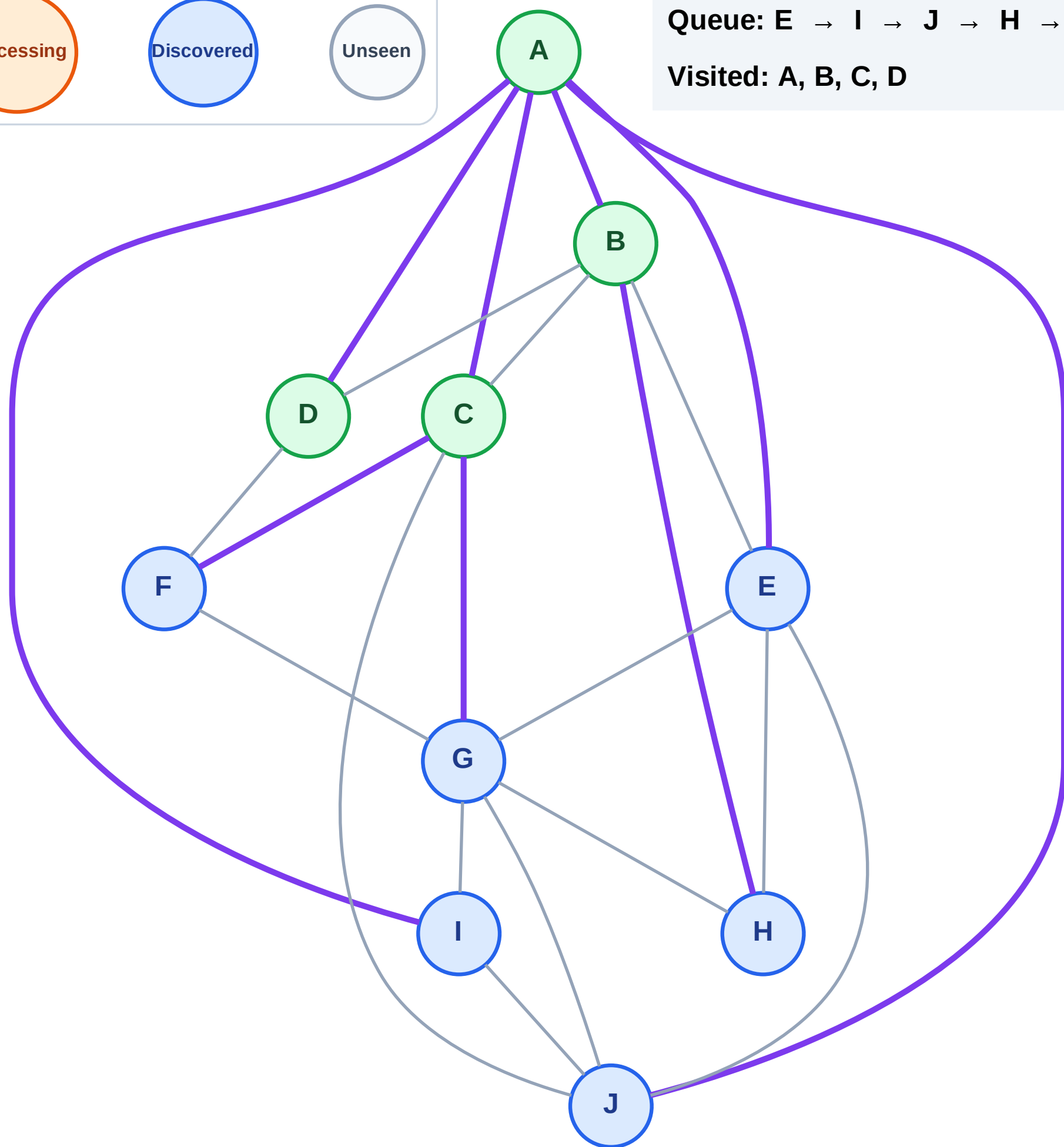
Discovered



Unseen

Queue: E → I → J → H → F → G

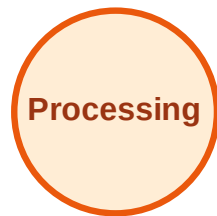
Visited: A, B, C, D



BFS Traversal

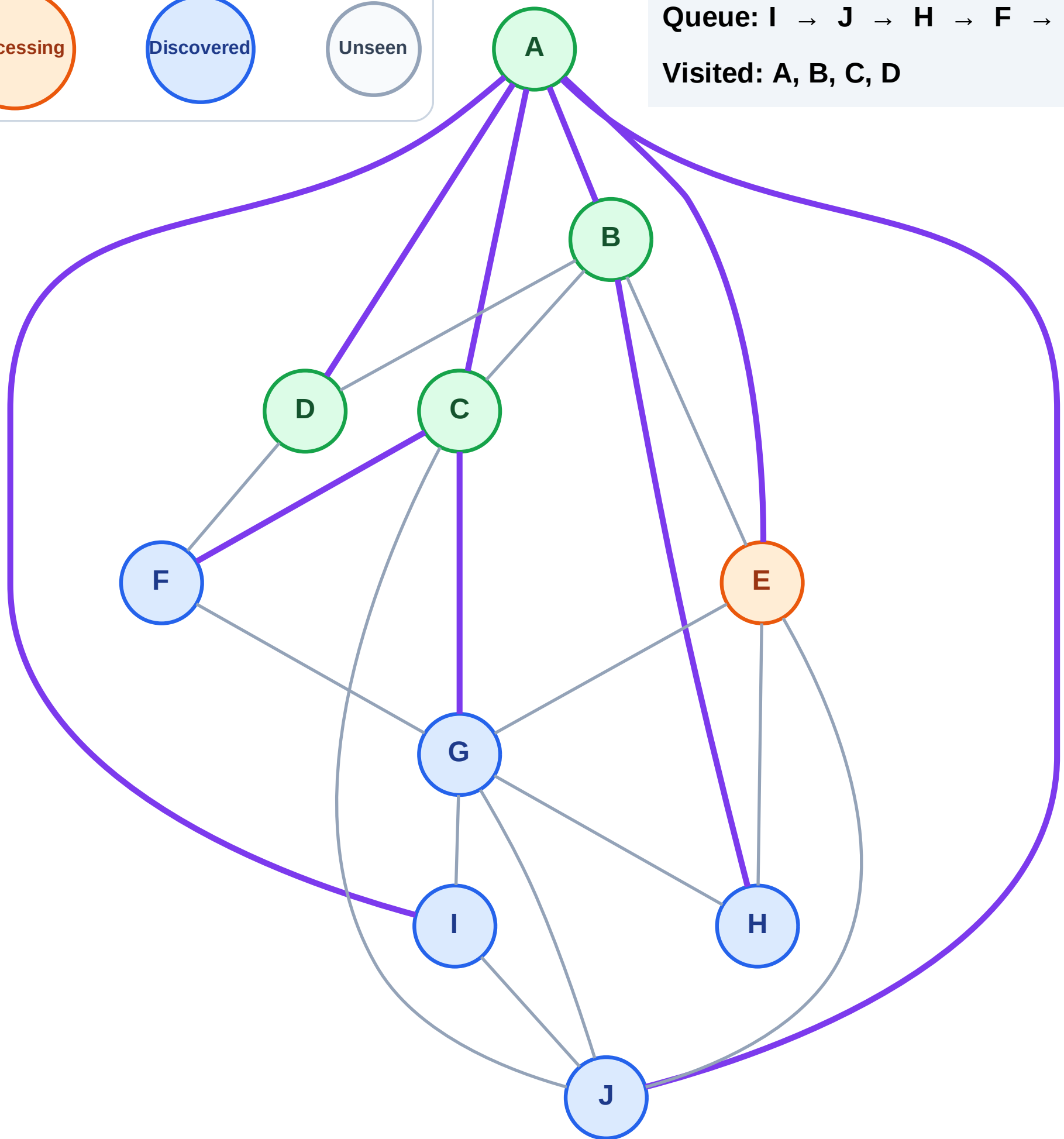
Step 10: Process node E

Legend



Queue: I → J → H → F → G

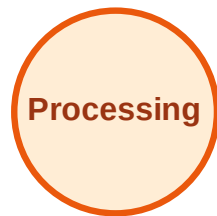
Visited: A, B, C, D



BFS Traversal

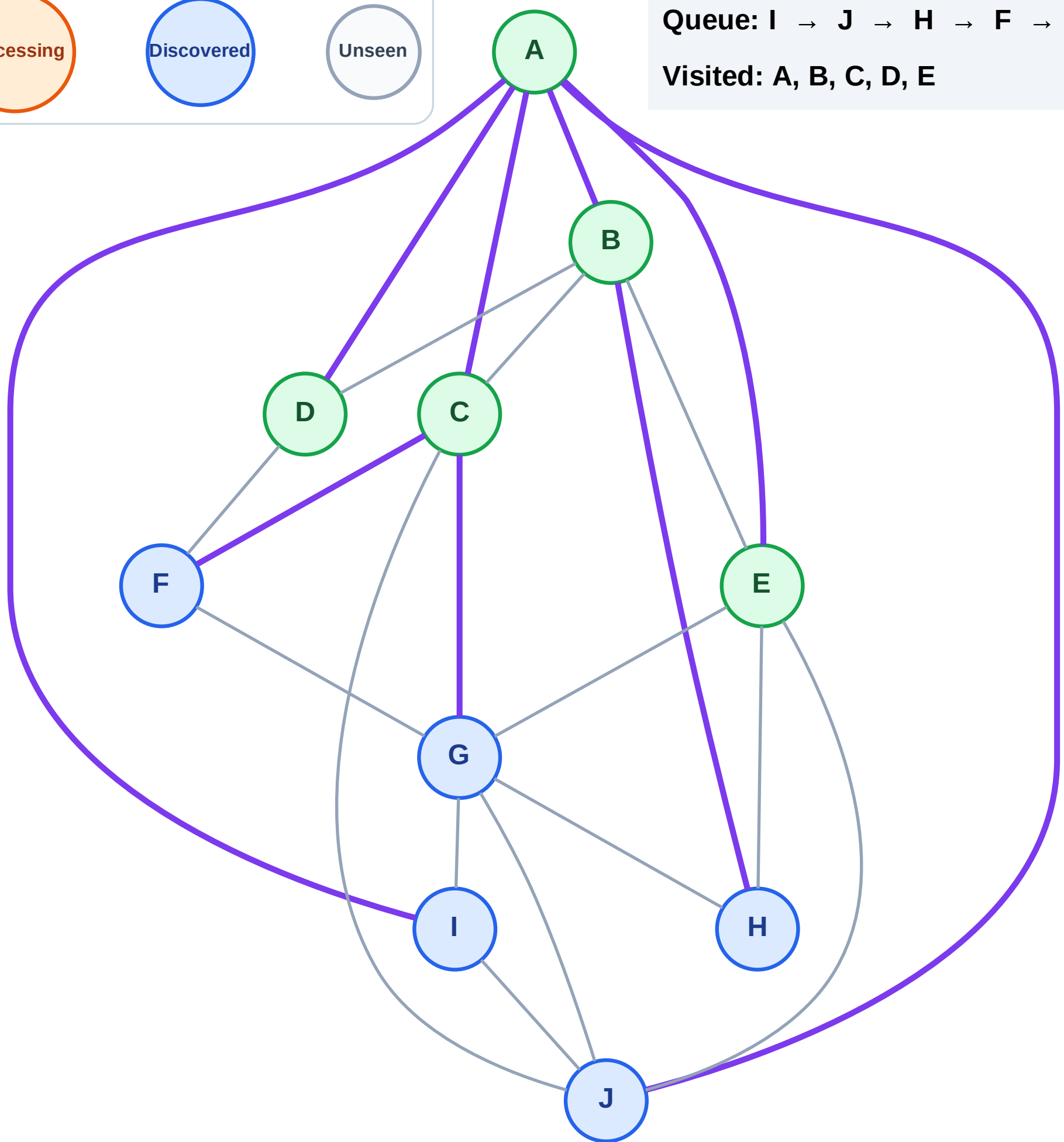
Step 11: No new neighbors from E

Legend



Queue: I → J → H → F → G

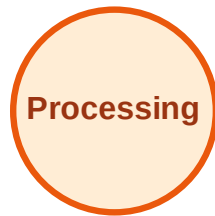
Visited: A, B, C, D, E



BFS Traversal

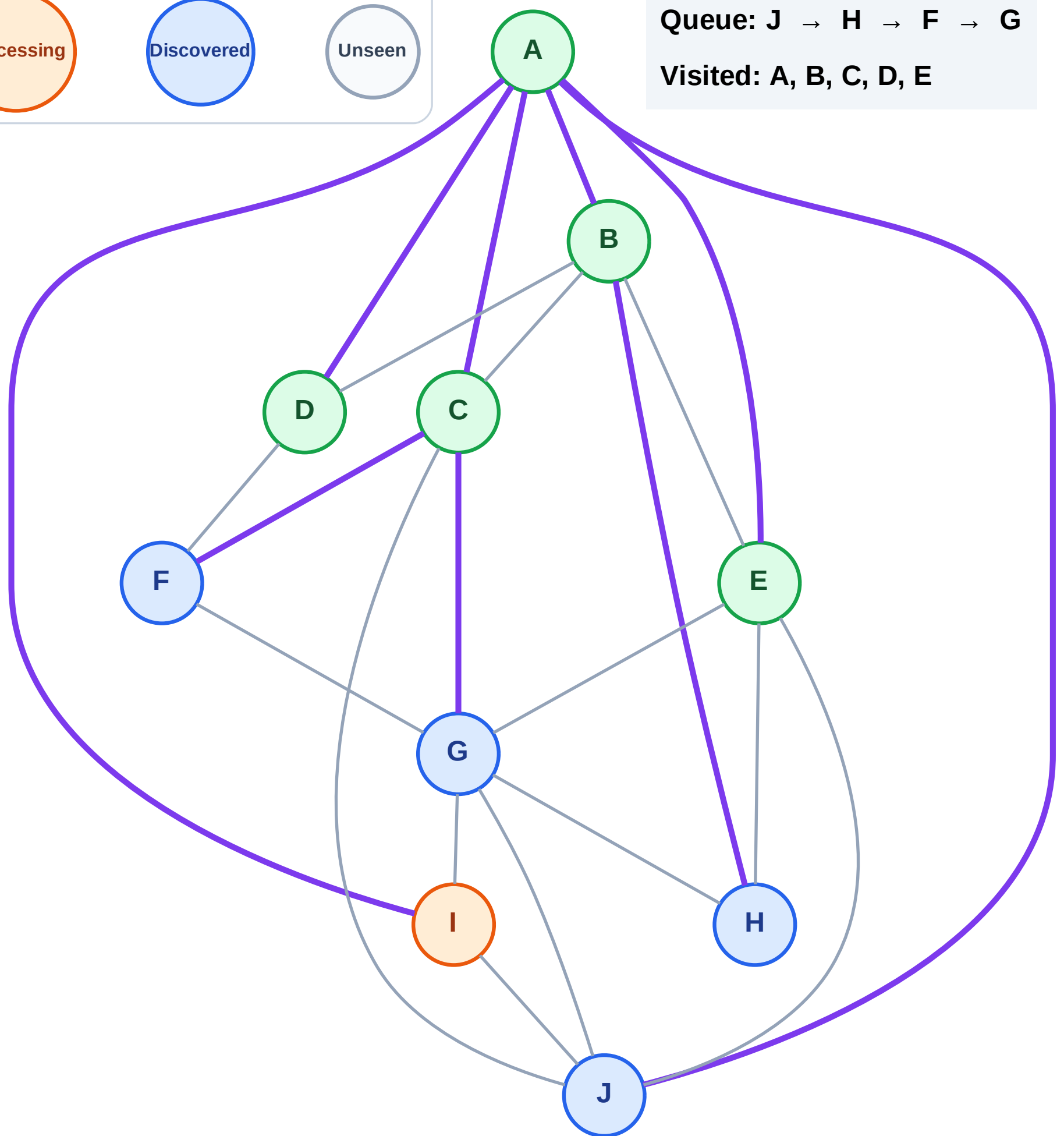
Step 12: Process node I

Legend



Queue: J → H → F → G

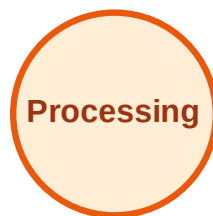
Visited: A, B, C, D, E



BFS Traversal

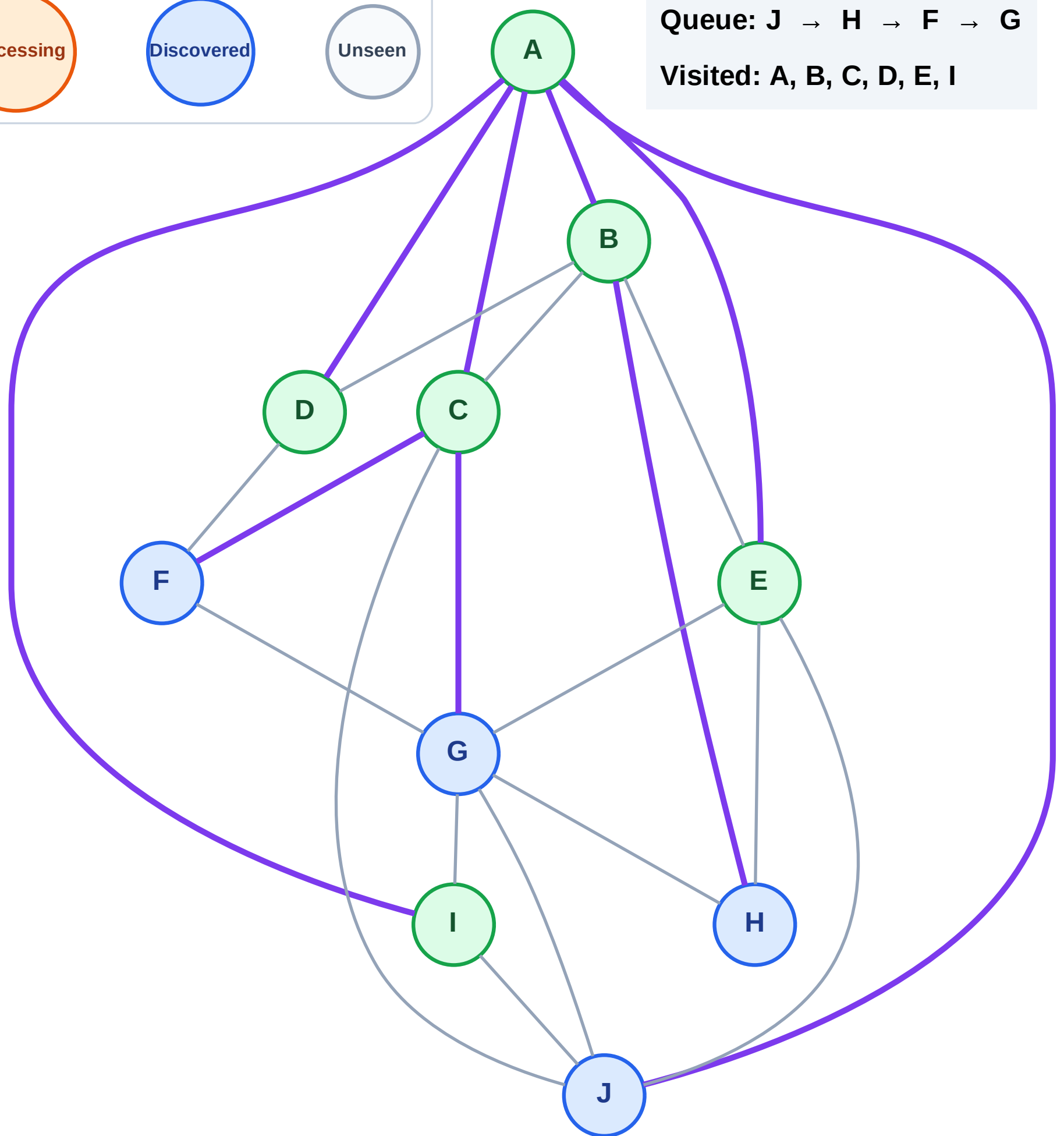
Step 13: No new neighbors from I

Legend



Queue: J → H → F → G

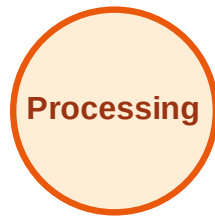
Visited: A, B, C, D, E, I



BFS Traversal

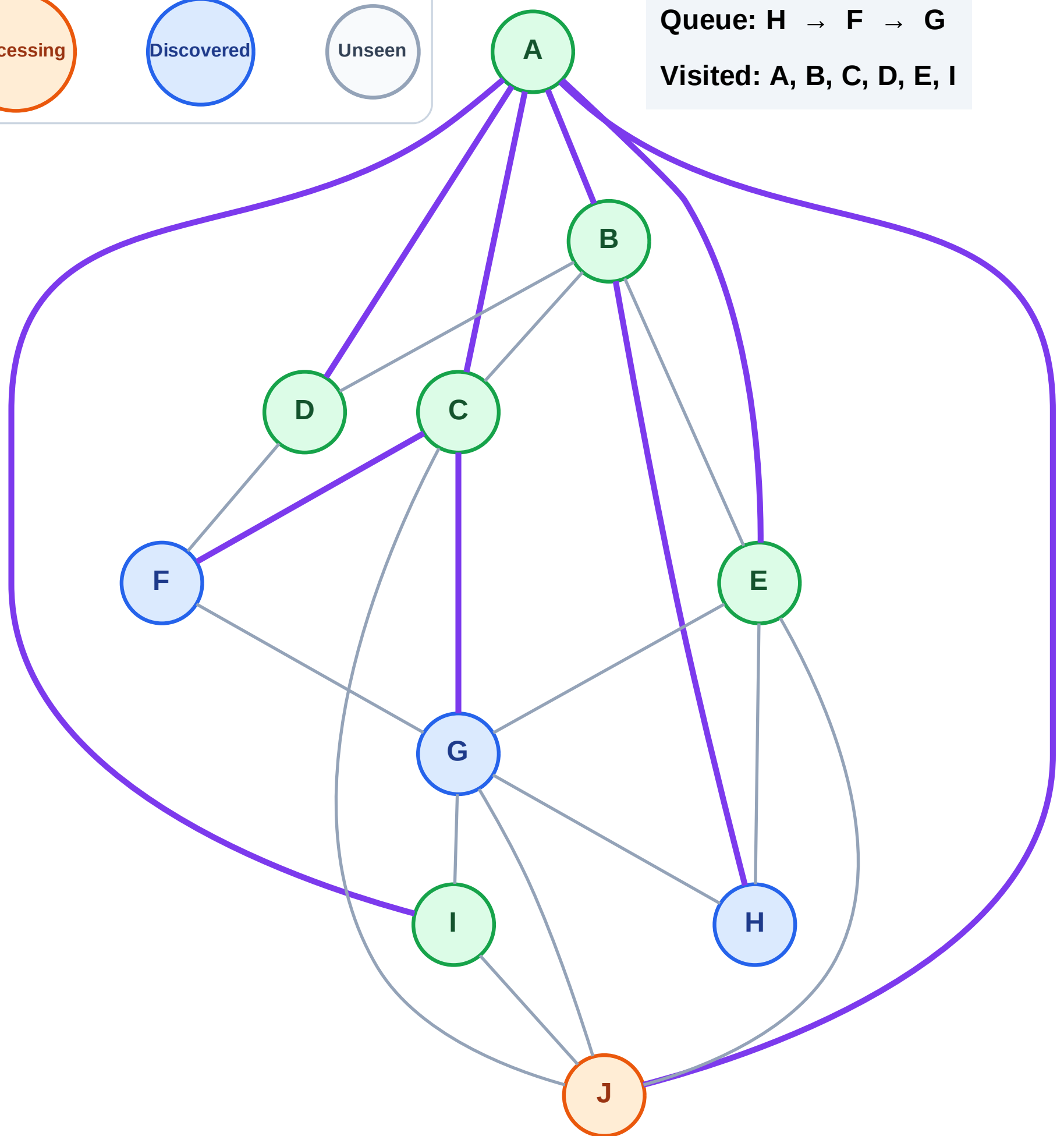
Step 14: Process node J

Legend



Queue: H → F → G

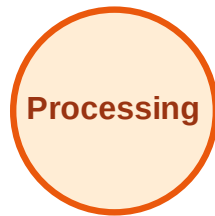
Visited: A, B, C, D, E, I



BFS Traversal

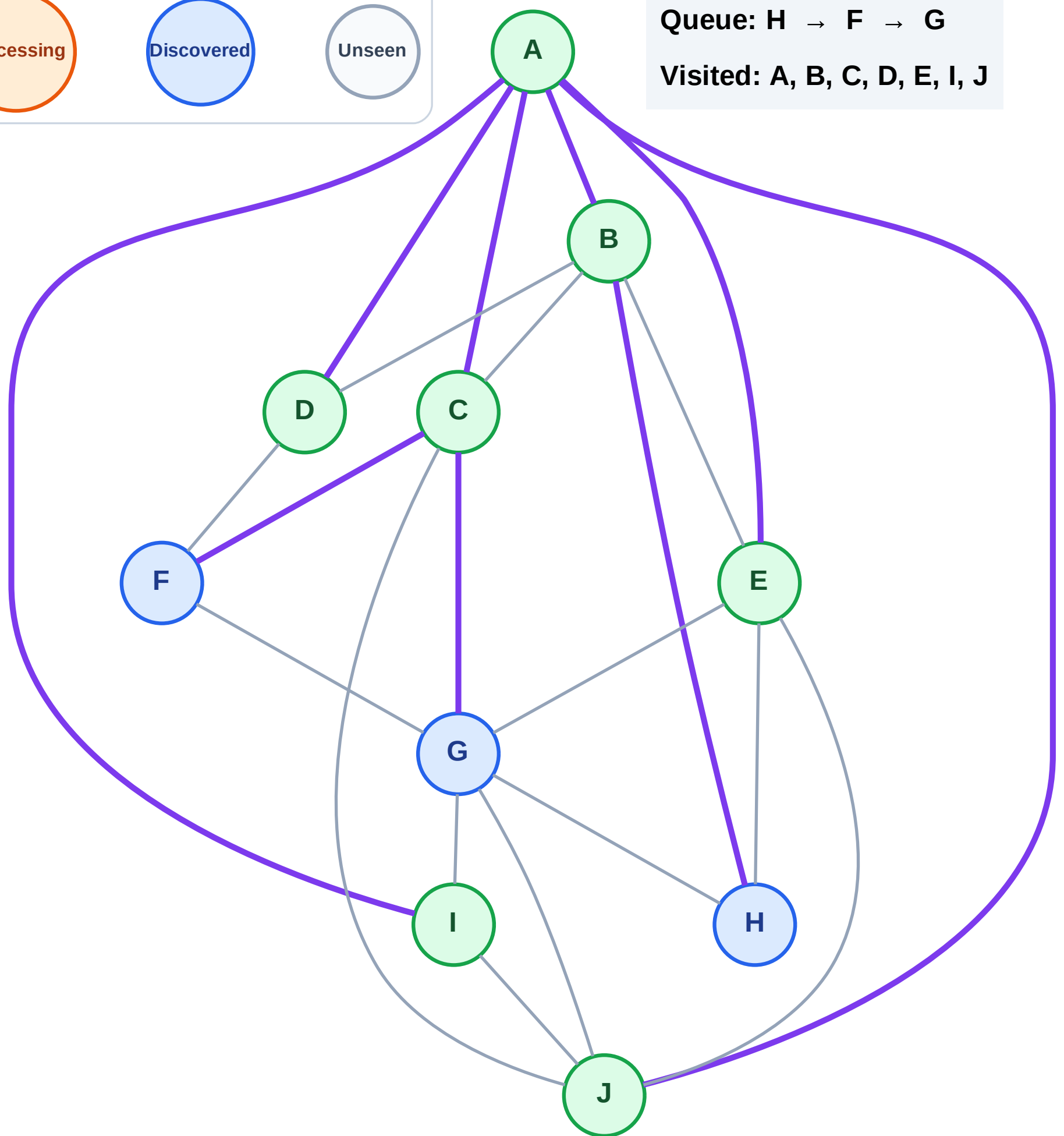
Step 15: No new neighbors from J

Legend



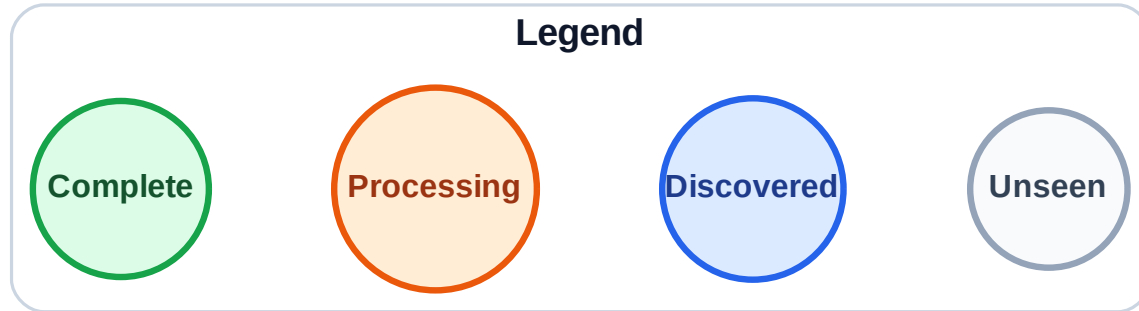
Queue: H → F → G

Visited: A, B, C, D, E, I, J



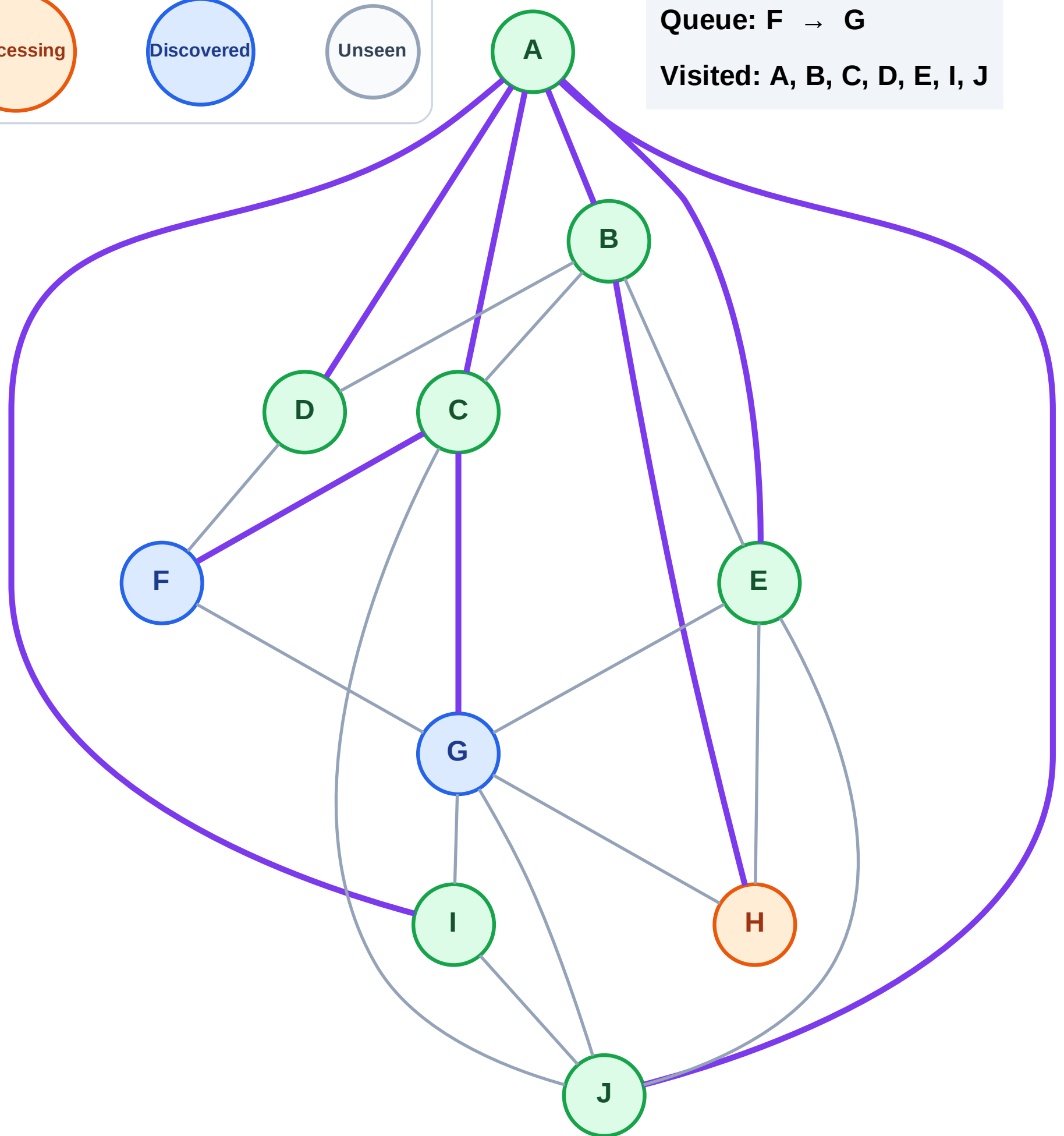
Step 16: Process node H

Step 16: Process node H



Queue: F → G

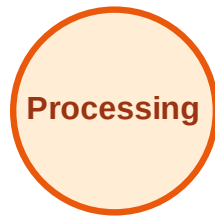
Visited: A, B, C, D, E, I, J



BFS Traversal

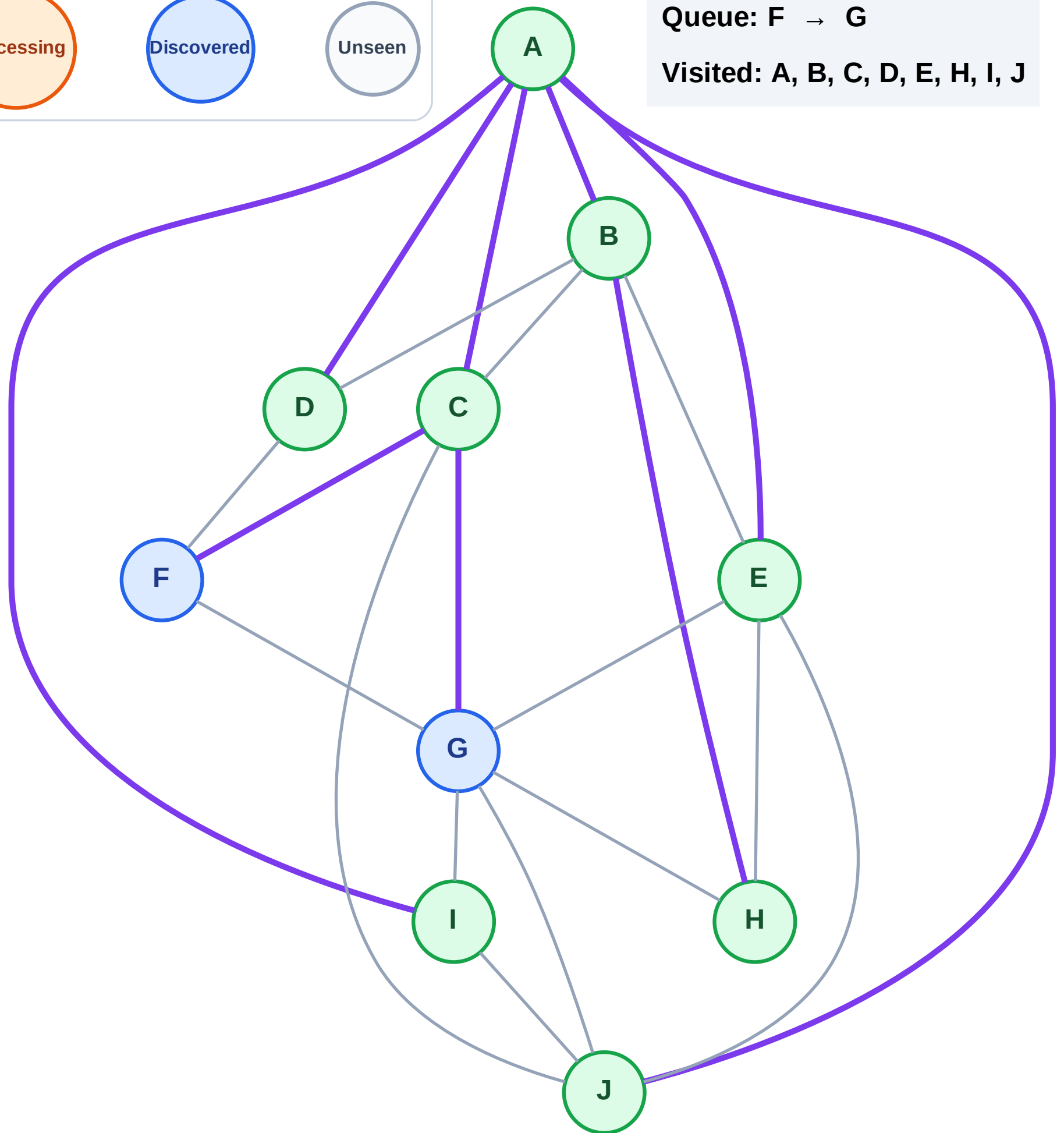
Step 17: No new neighbors from H

Legend

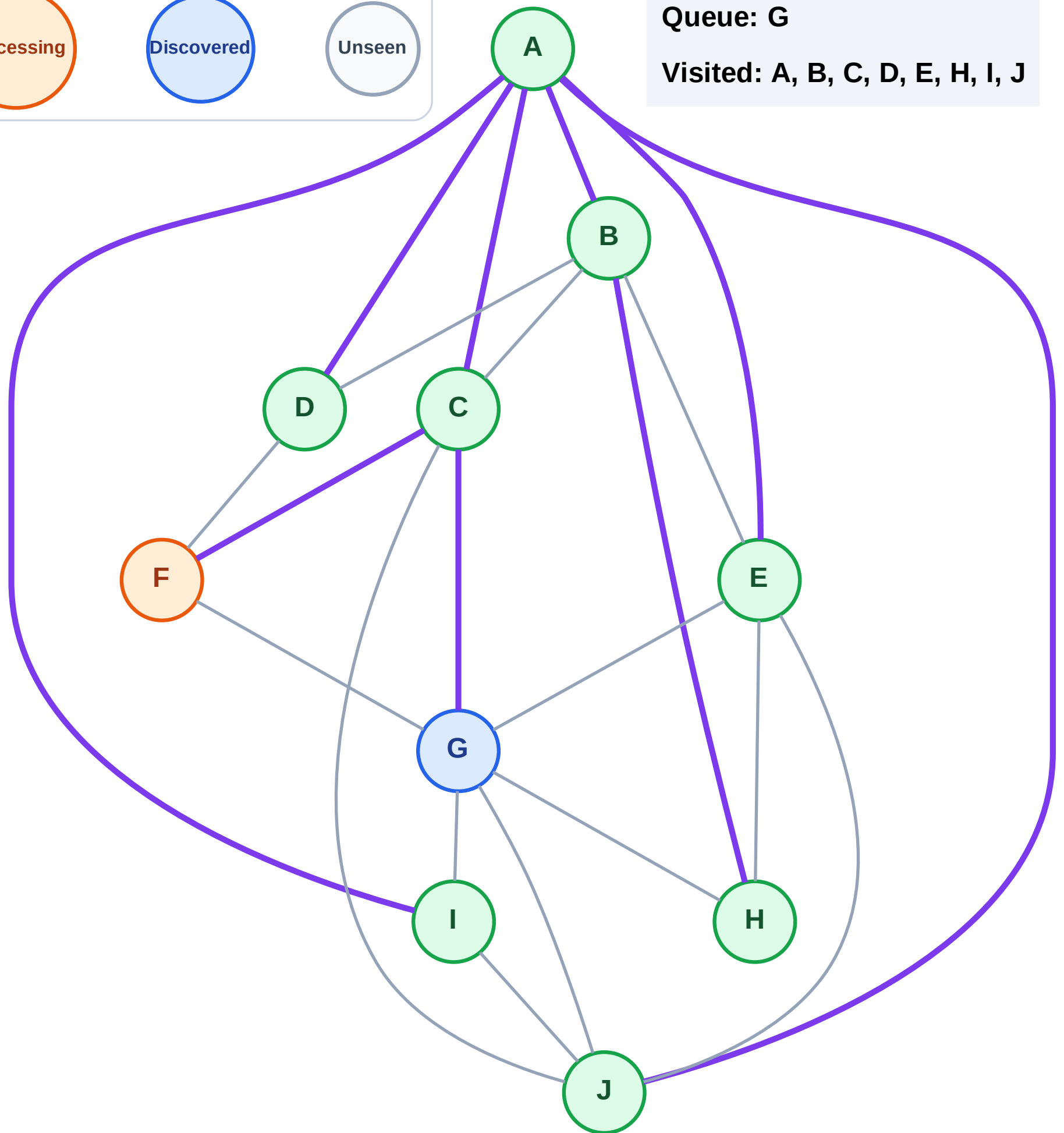
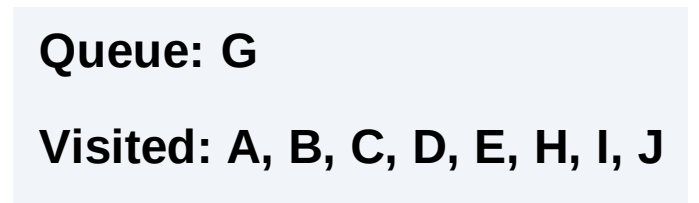


Queue: F → G

Visited: A, B, C, D, E, H, I, J



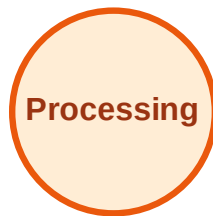
Step 18: Process node F



BFS Traversal

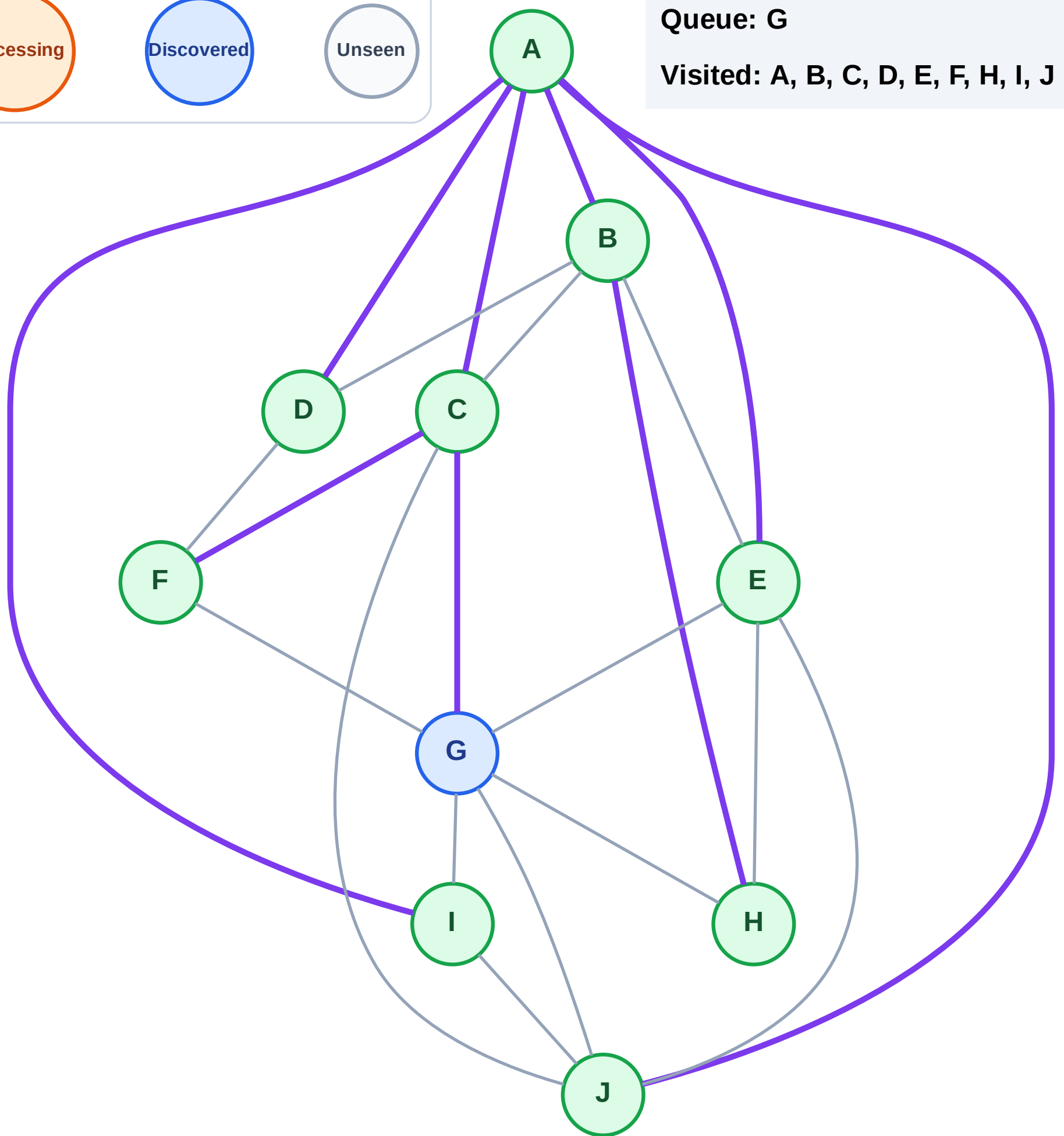
Step 19: No new neighbors from F

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

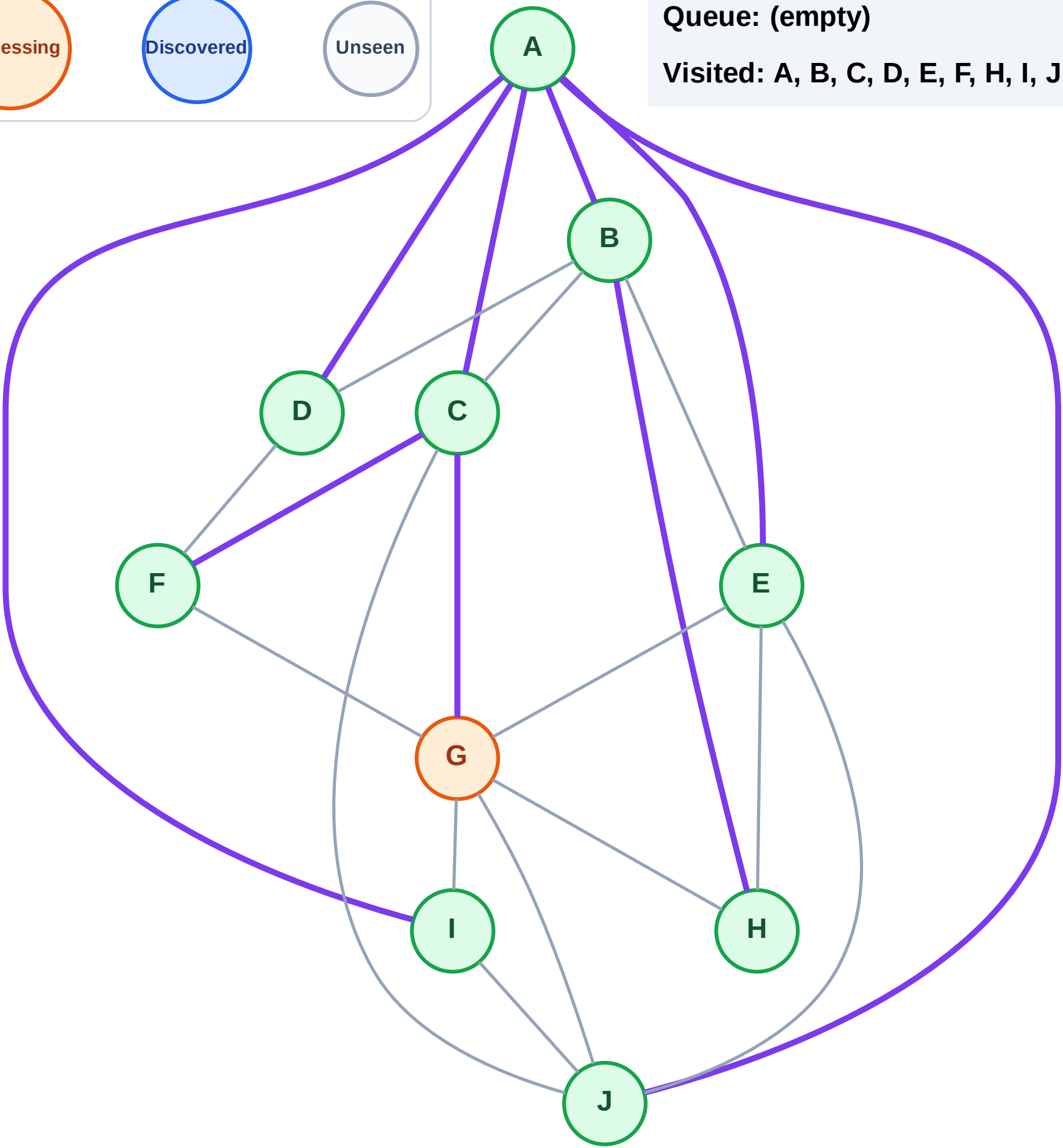
Step 20: Process node G

Legend



Queue: (empty)

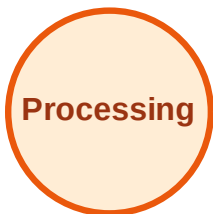
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

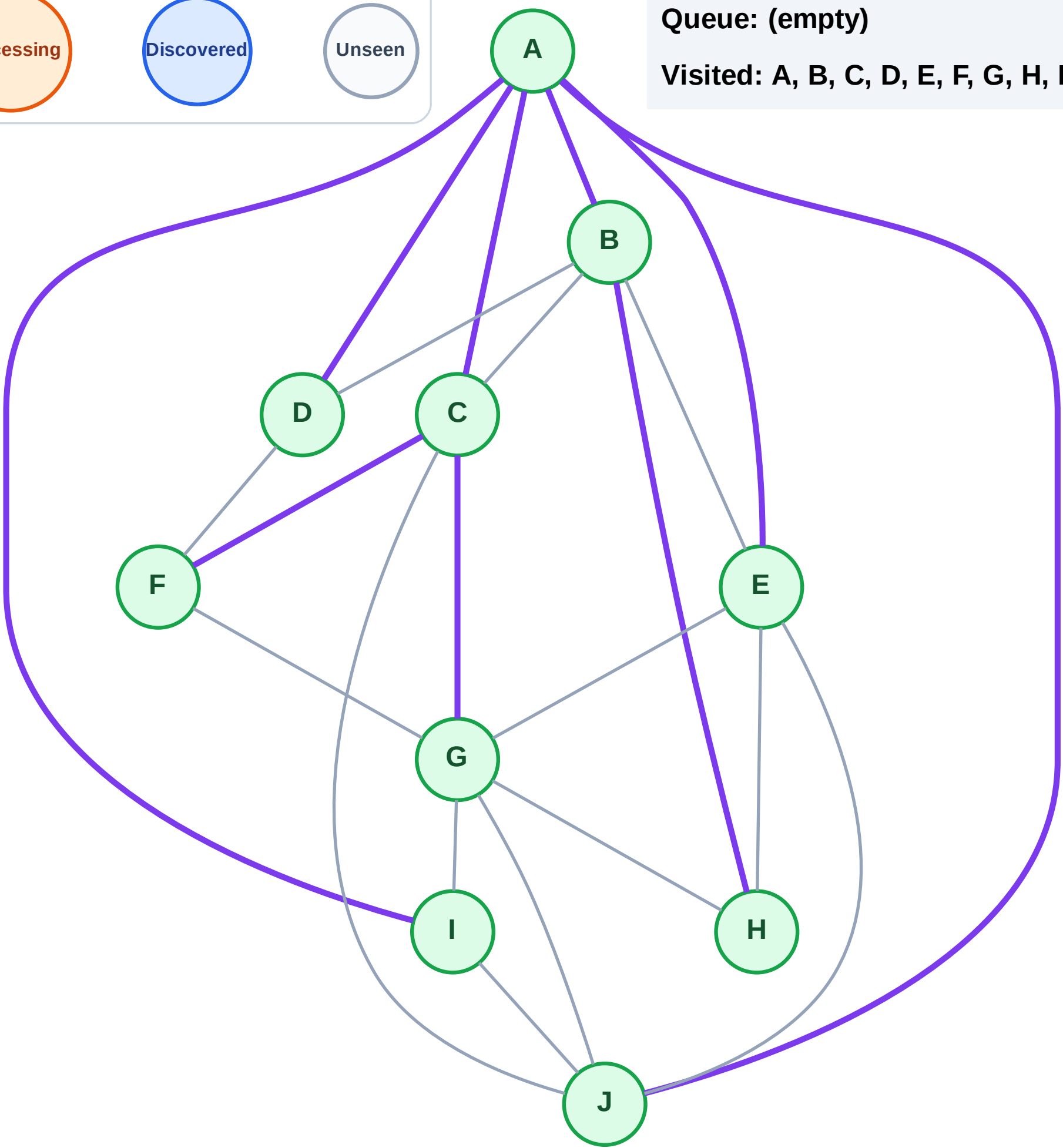
Step 21: No new neighbors from G

Legend



Queue: (empty)

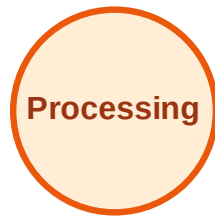
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

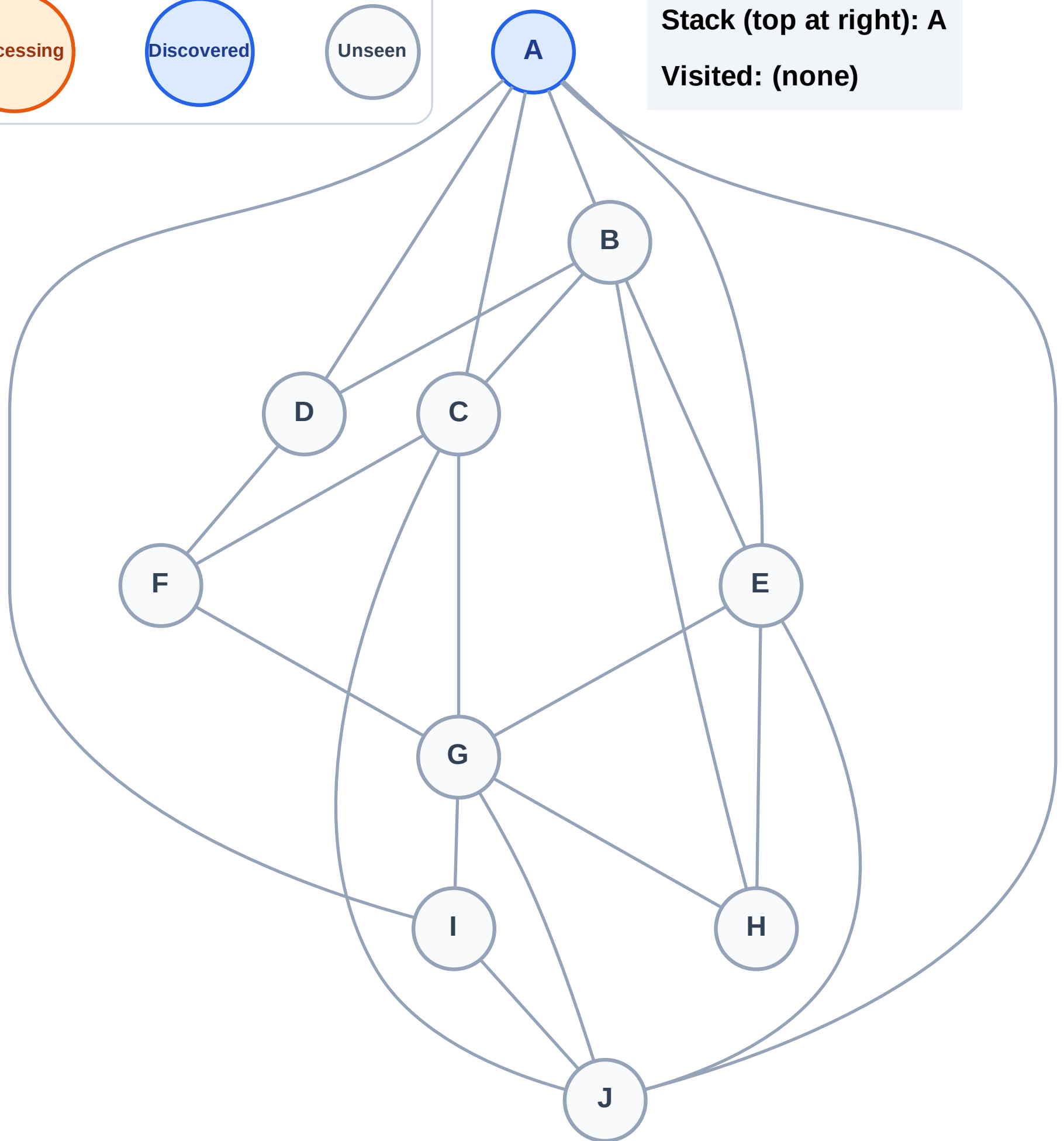
Step 1: Start at A

Legend



Stack (top at right): A

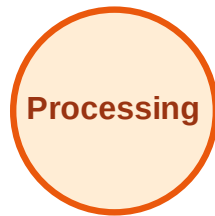
Visited: (none)



DFS Traversal

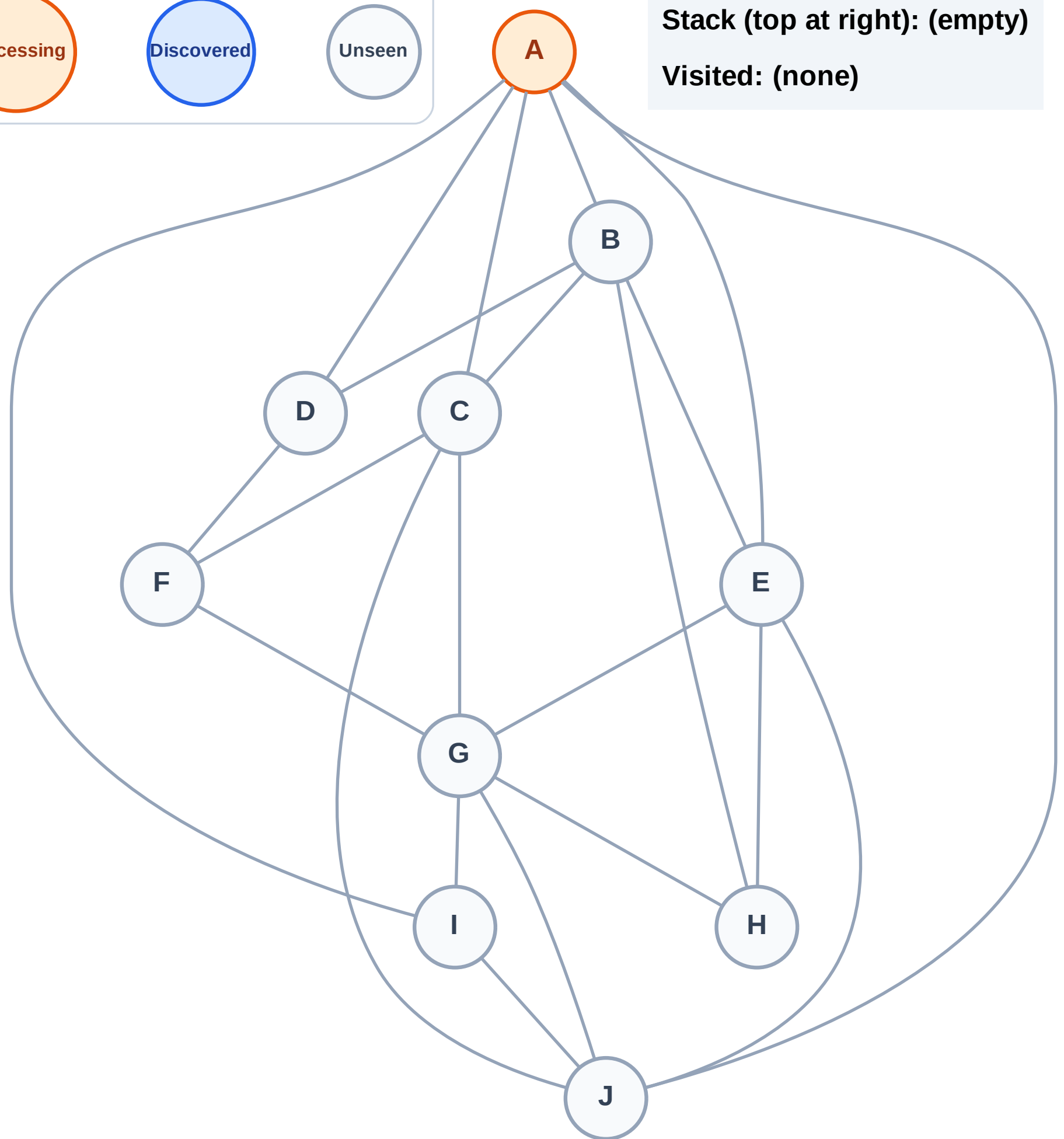
Step 2: Process node A

Legend



Stack (top at right): (empty)

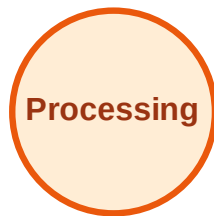
Visited: (none)



DFS Traversal

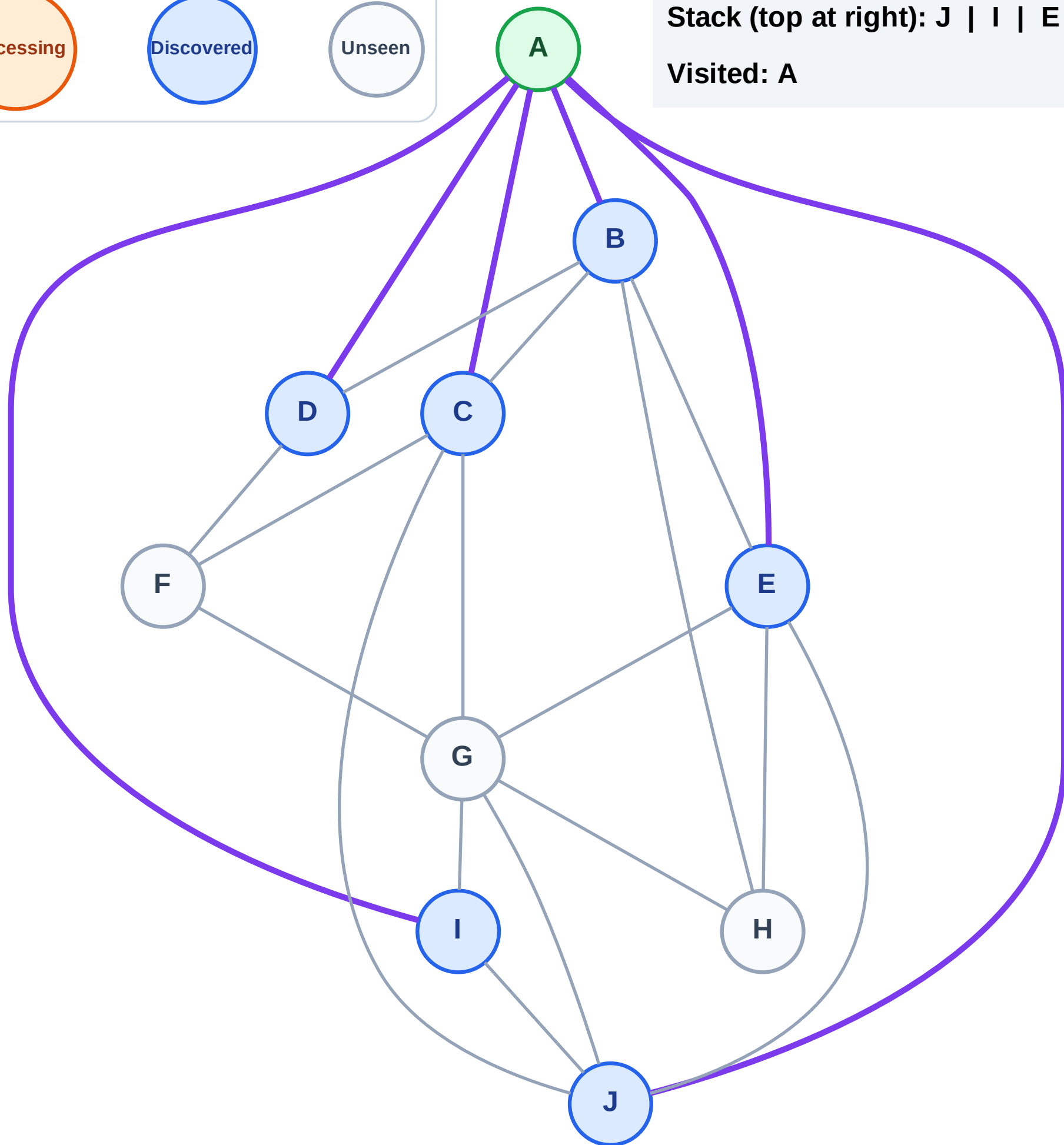
Step 3: Discover J, I, E, D, C, B from A

Legend



Stack (top at right): J | I | E | D | C | B

Visited: A



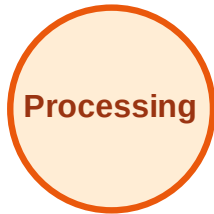
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



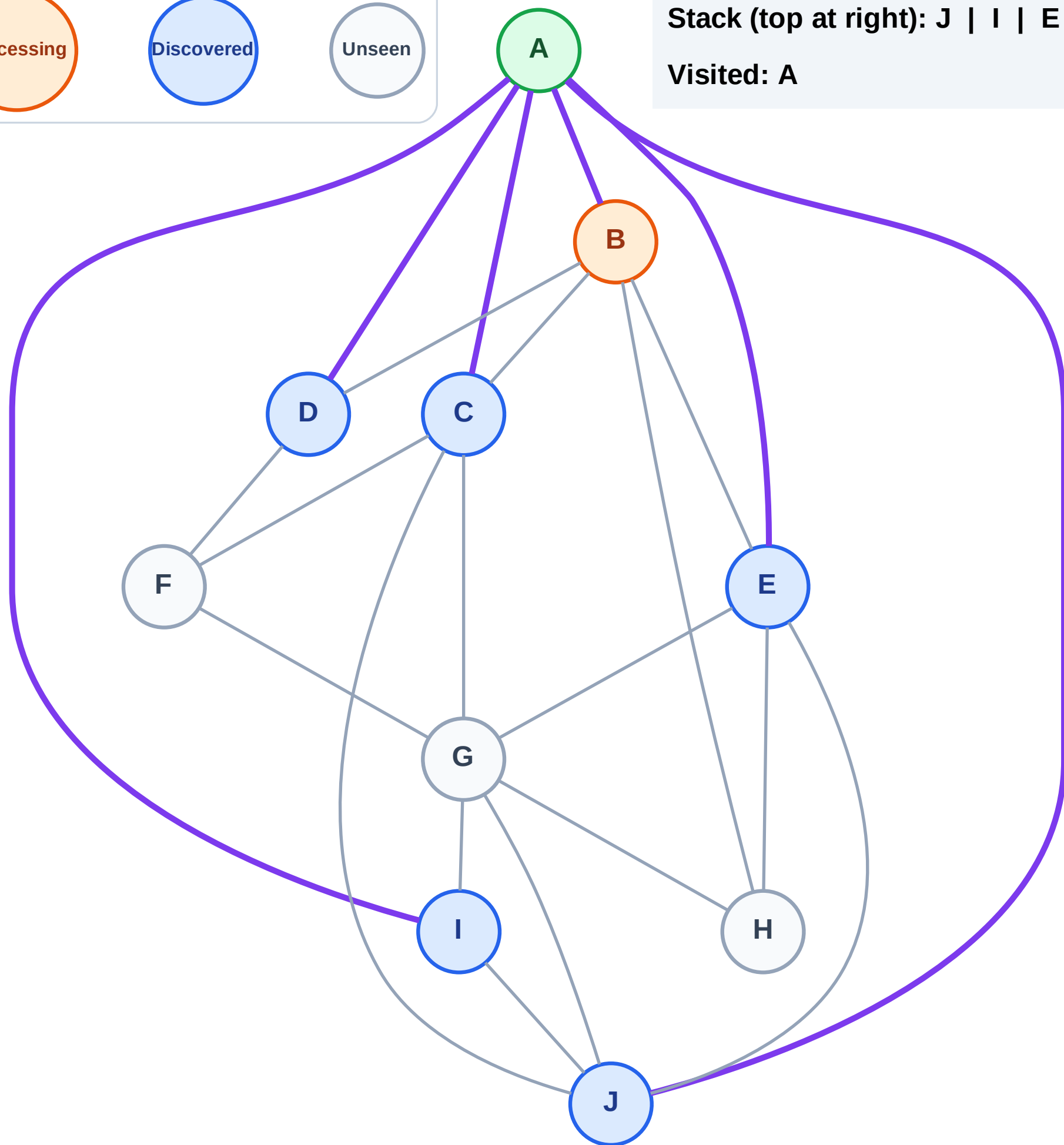
Discovered



Unseen

Stack (top at right): J | I | E | D | C

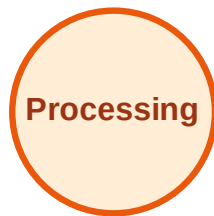
Visited: A



DFS Traversal

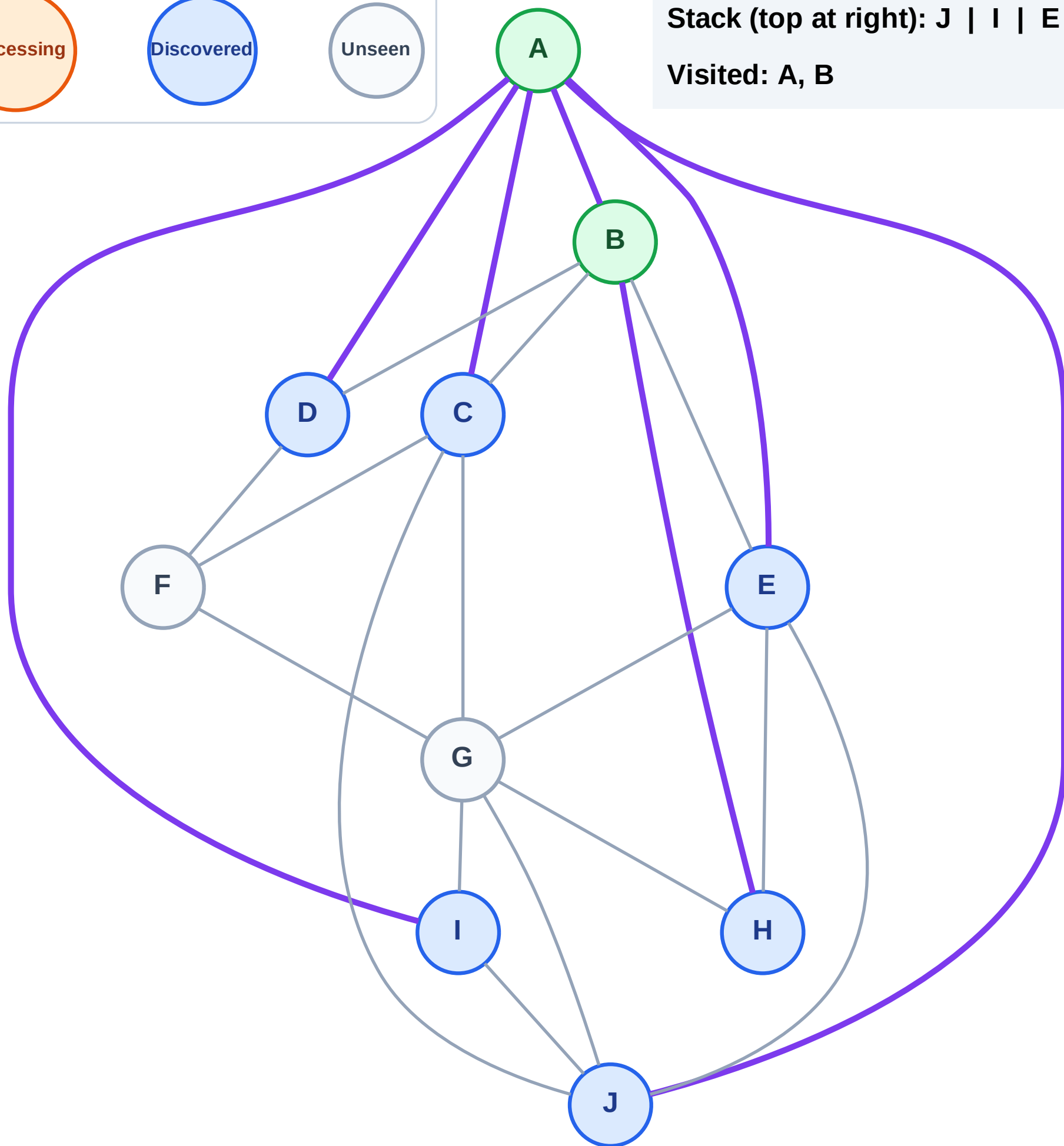
Step 5: Discover H from B

Legend



Stack (top at right): J | I | E | D | C | H

Visited: A, B



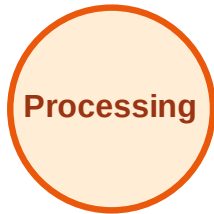
DFS Traversal

Step 6: Process node H

Legend



Complete



Processing



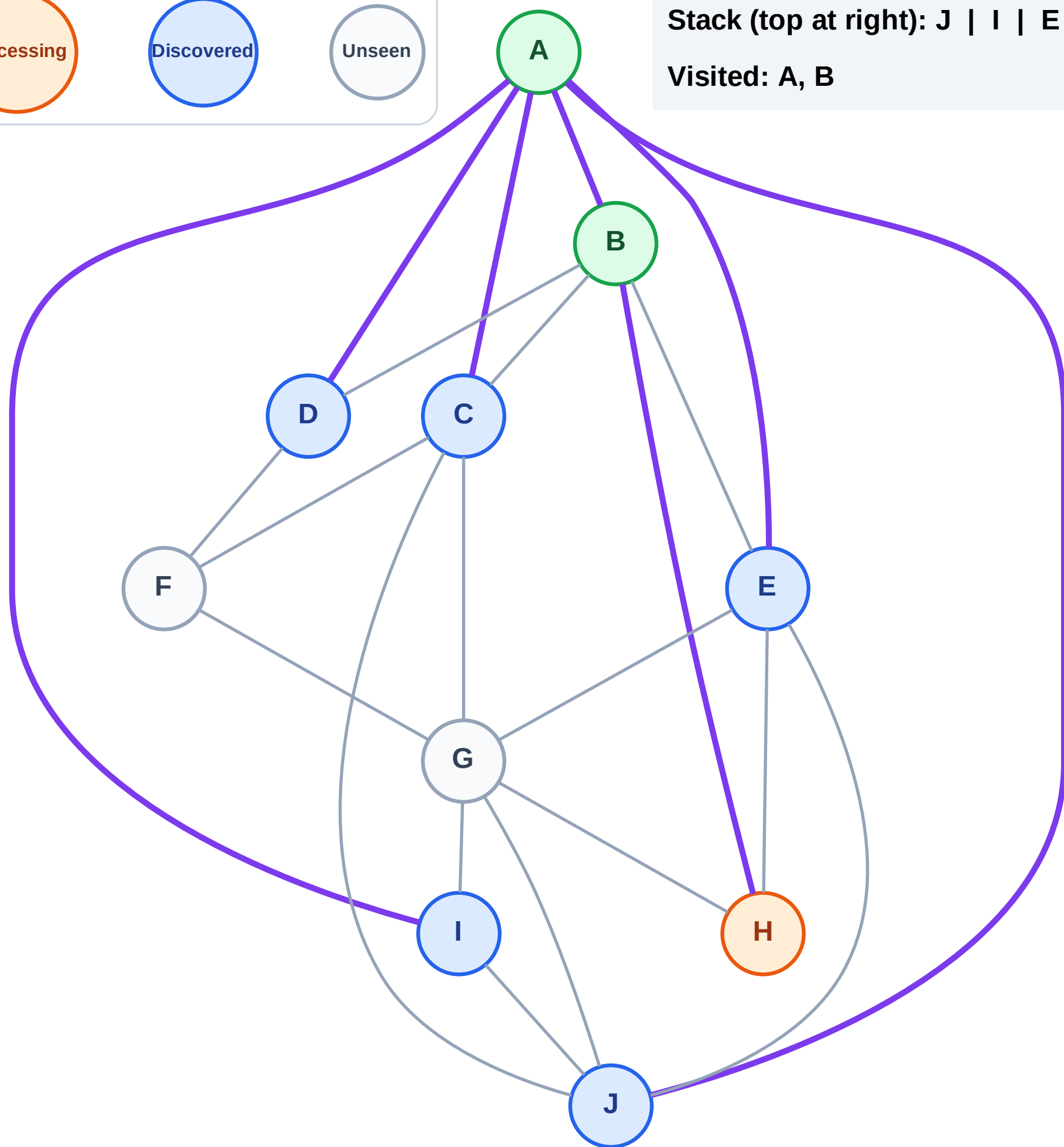
Discovered



Unseen

Stack (top at right): J | I | E | D | C

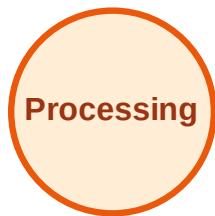
Visited: A, B



DFS Traversal

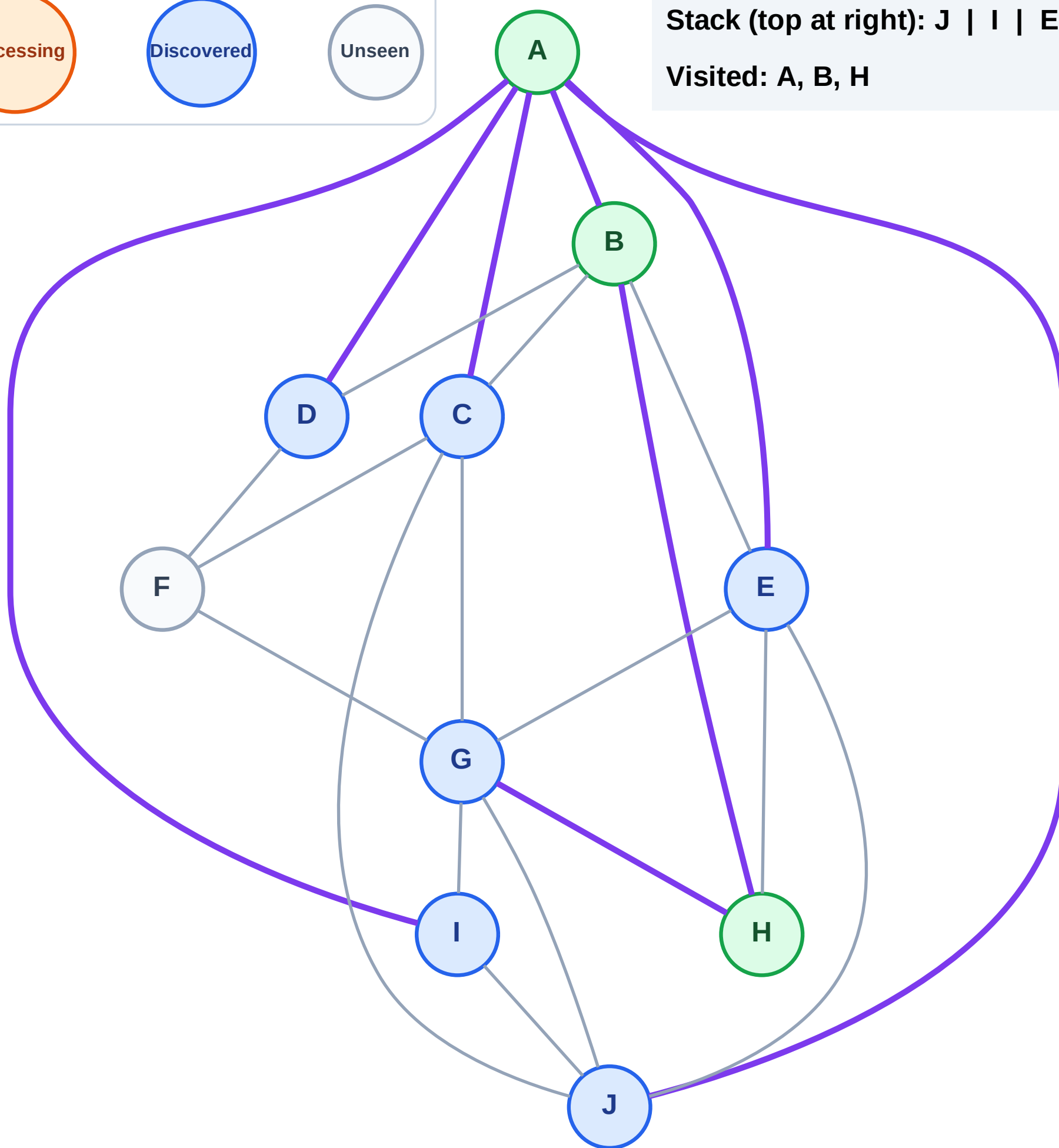
Step 7: Discover G from H

Legend

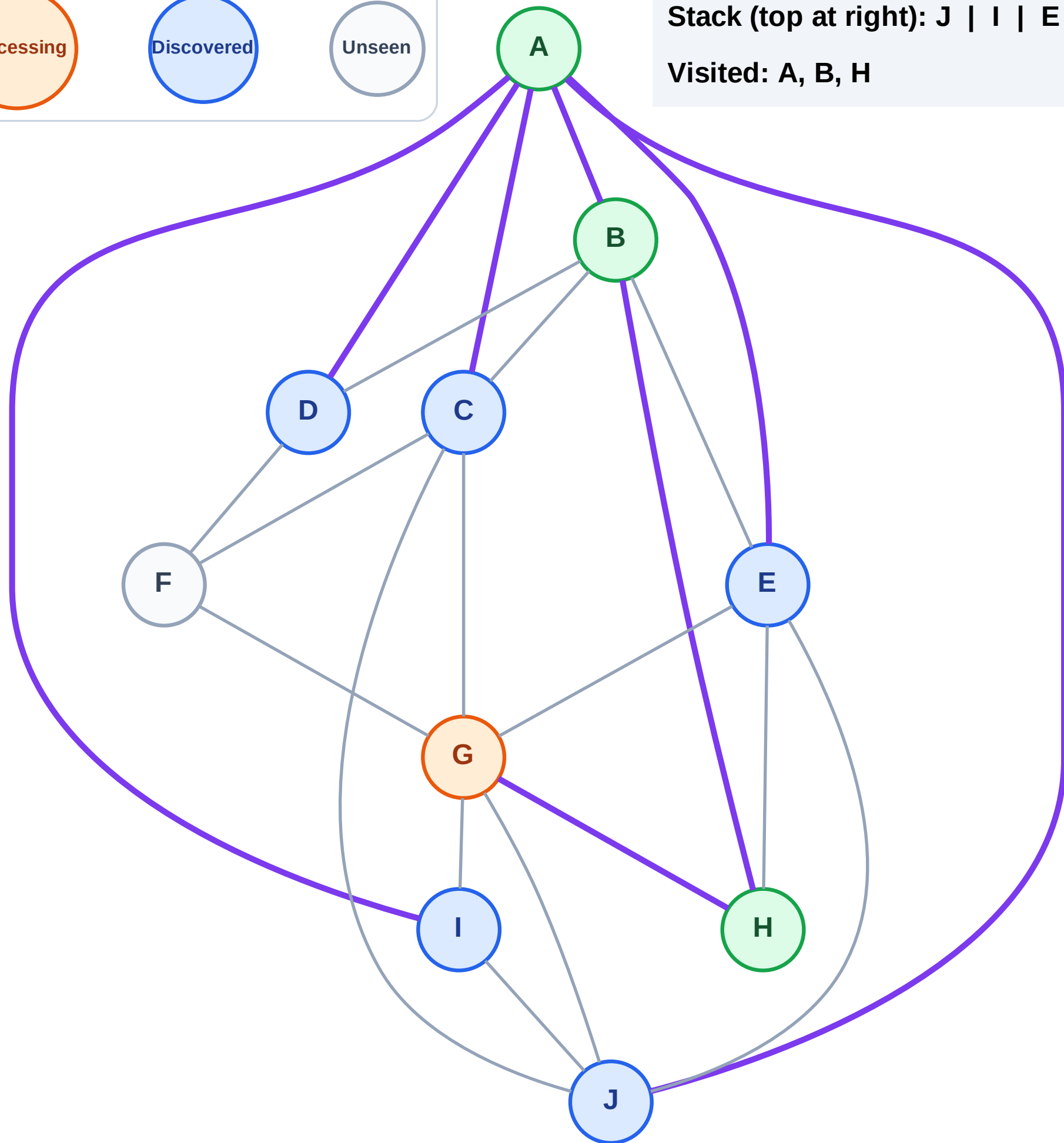
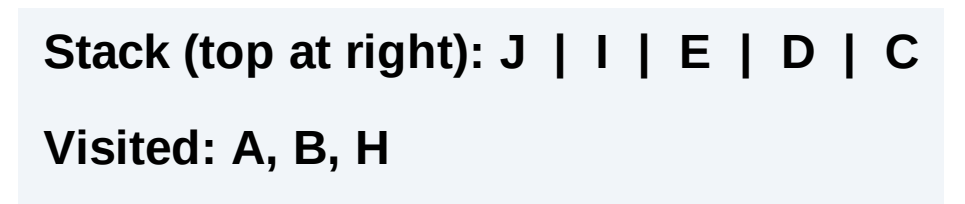


Stack (top at right): J | I | E | D | C | G

Visited: A, B, H



Step 8: Process node G



DFS Traversal

Step 9: Discover F from G

Legend

Complete

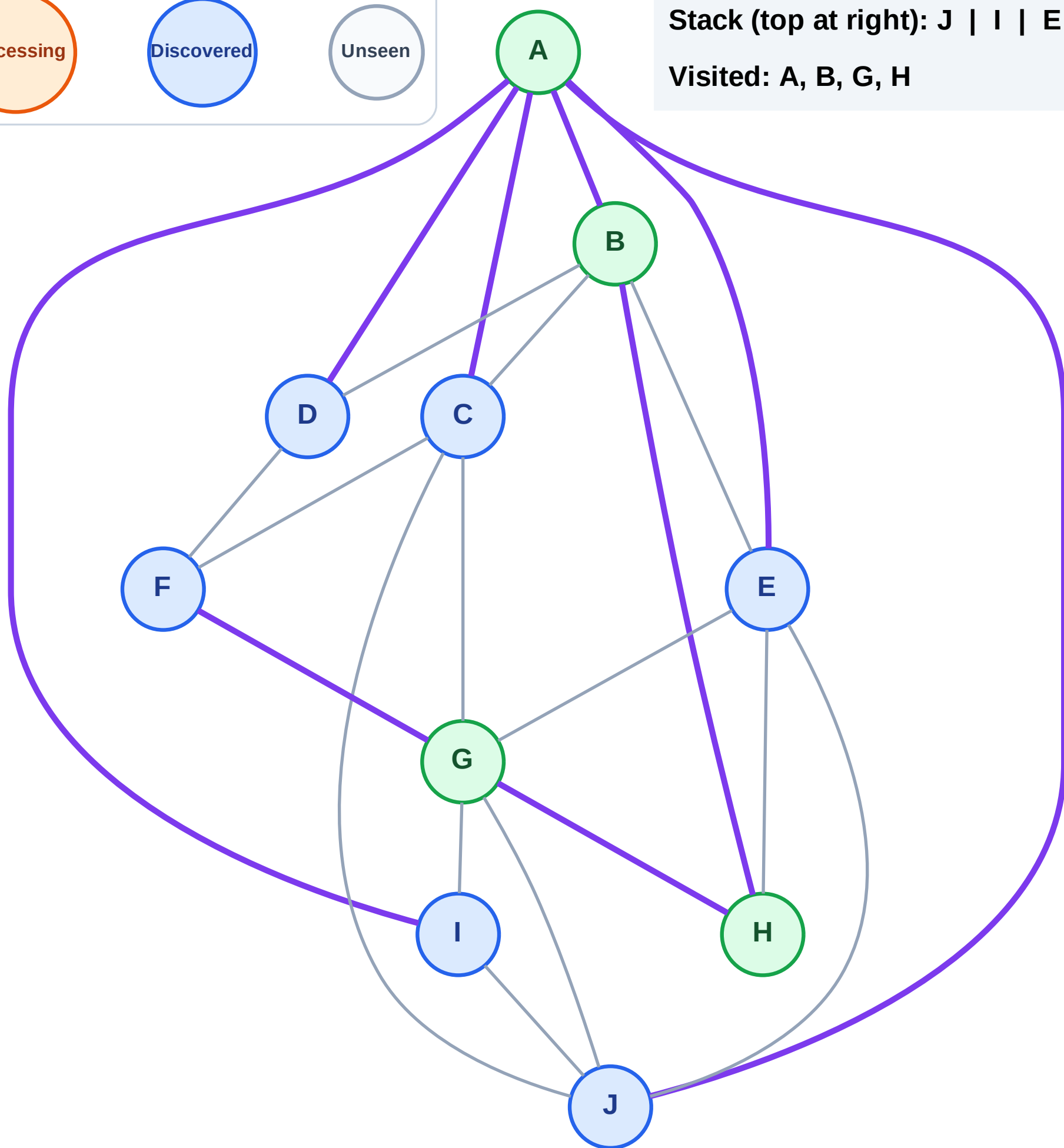
Processing

Discovered

Unseen

Stack (top at right): J | I | E | D | C | F

Visited: A, B, G, H



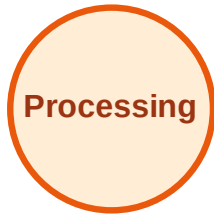
DFS Traversal

Step 10: Process node F

Legend



Complete



Processing



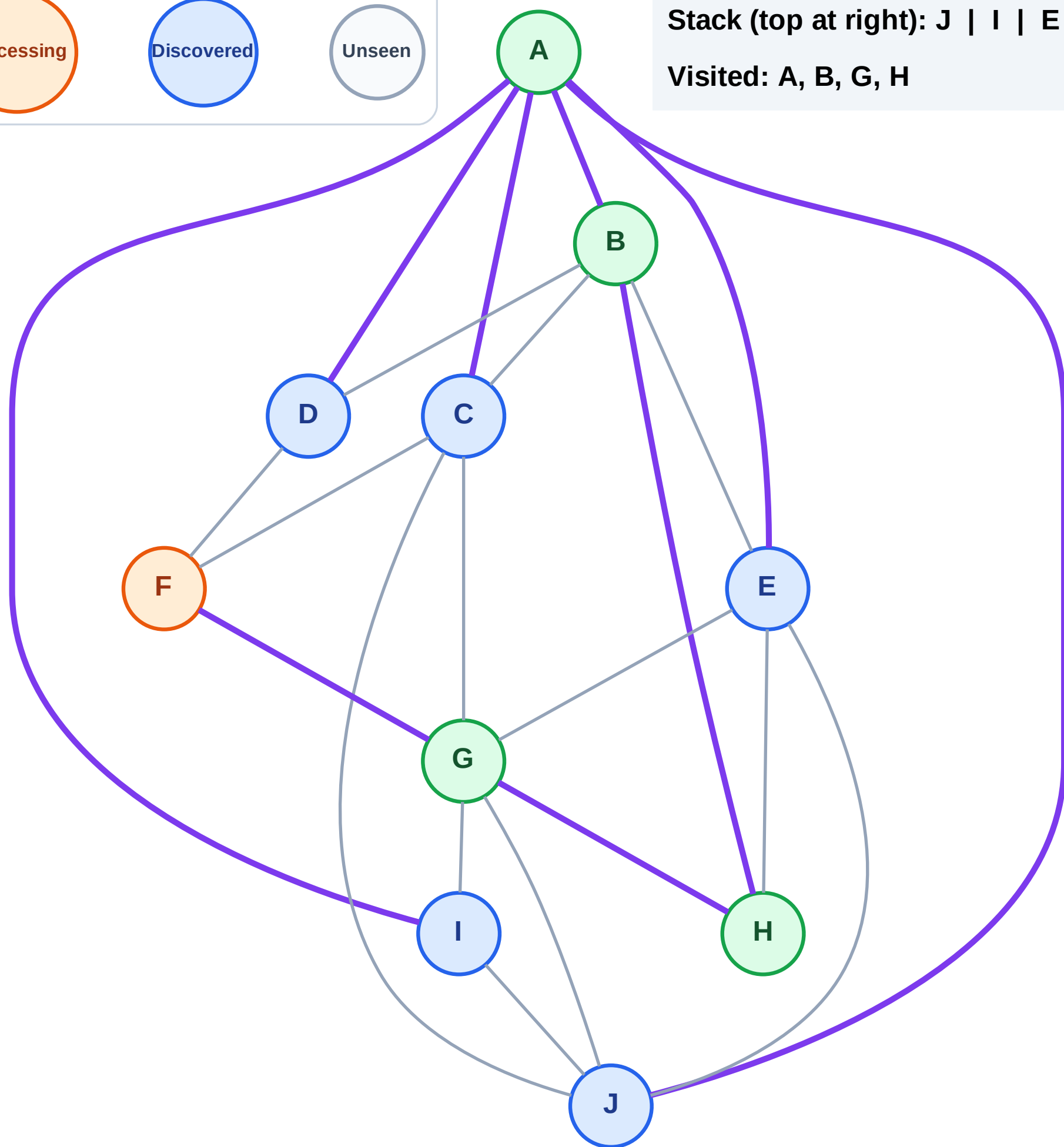
Discovered



Unseen

Stack (top at right): J | I | E | D | C

Visited: A, B, G, H



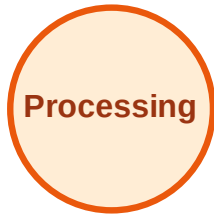
DFS Traversal

Step 11: No new neighbors from F

Legend



Complete



Processing



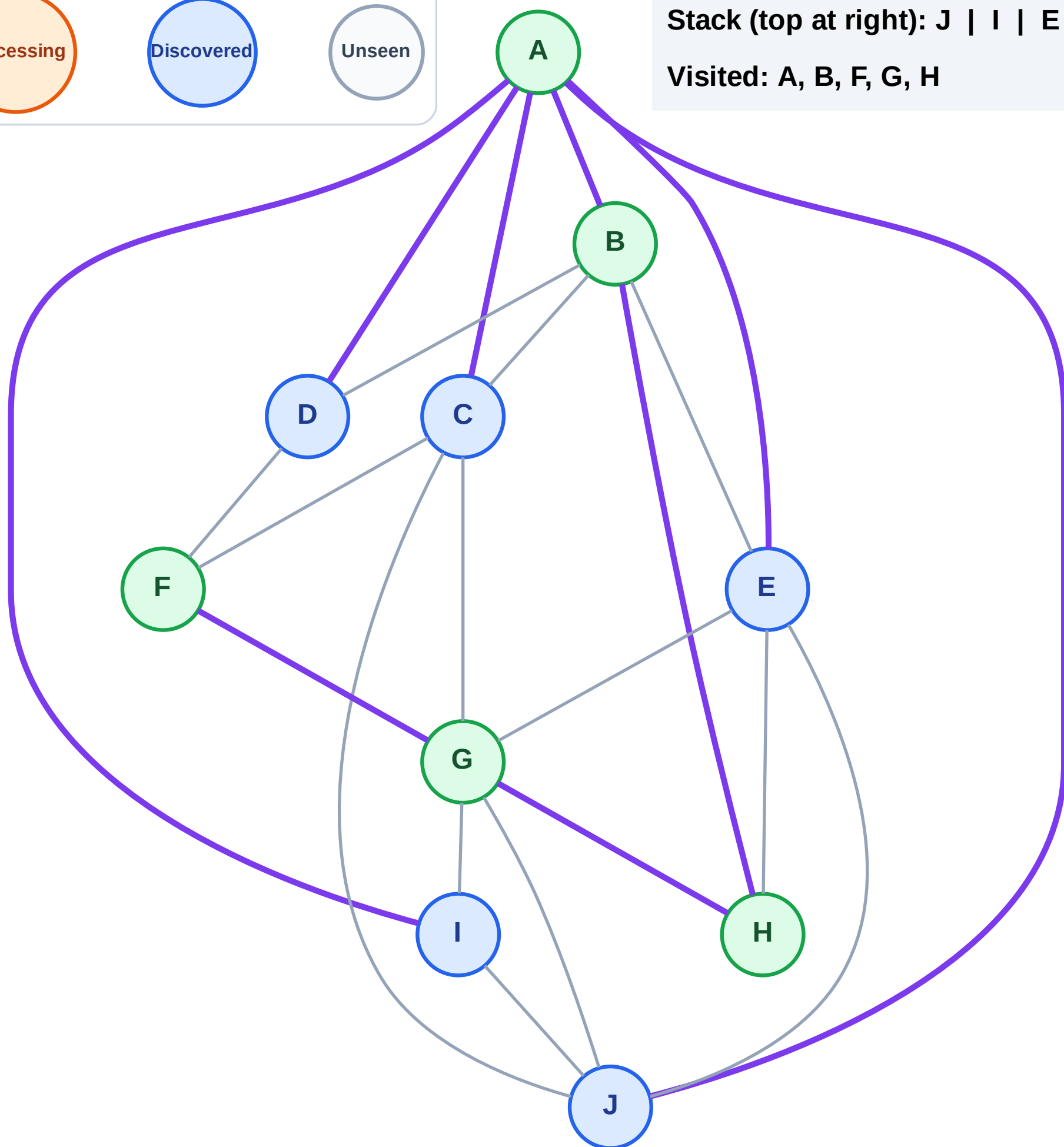
Discovered



Unseen

Stack (top at right): J | I | E | D | C

Visited: A, B, F, G, H



DFS Traversal

Step 12: Process node C

Legend



Complete



Processing



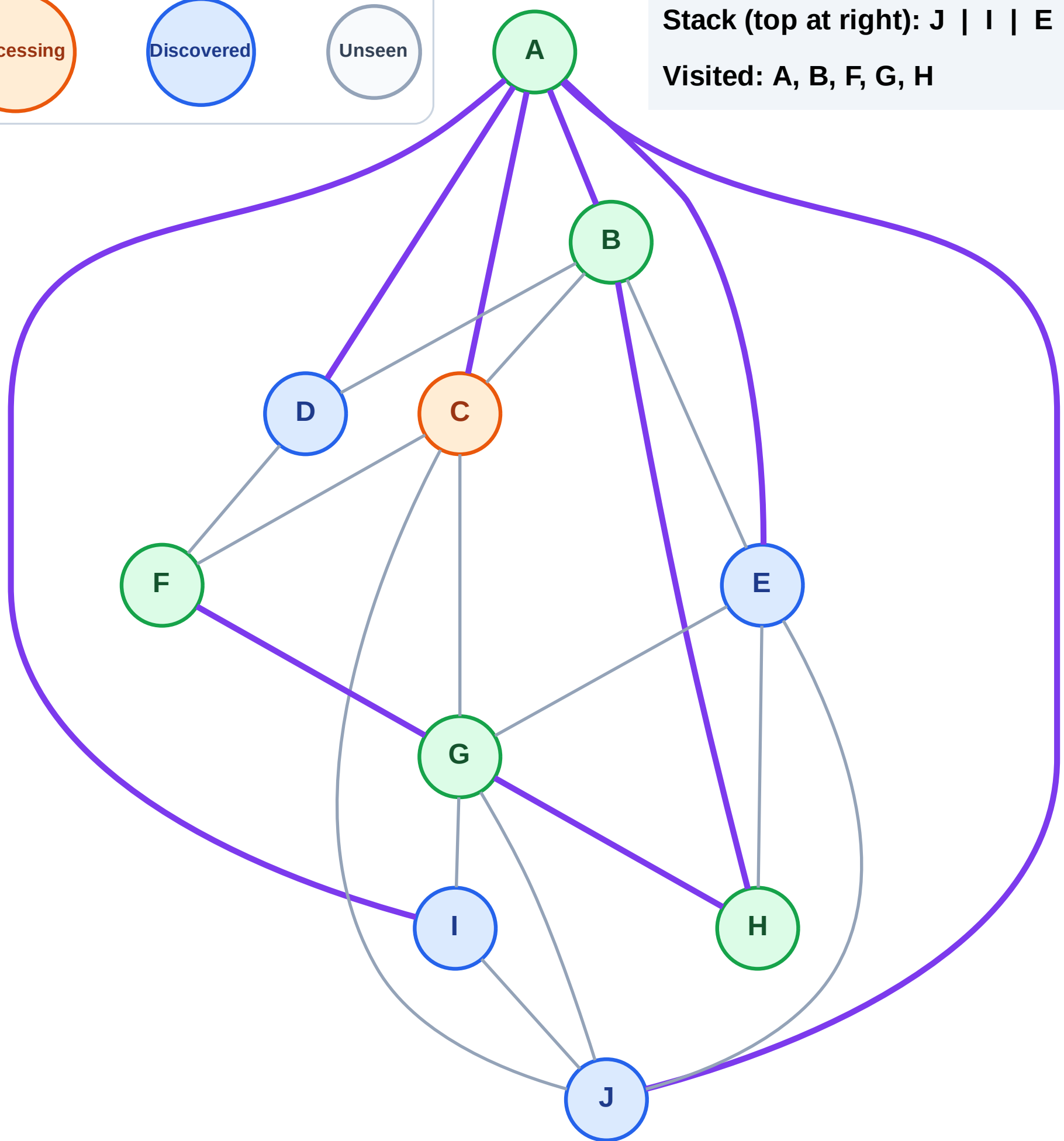
Discovered



Unseen

Stack (top at right): J | I | E | D

Visited: A, B, F, G, H



DFS Traversal

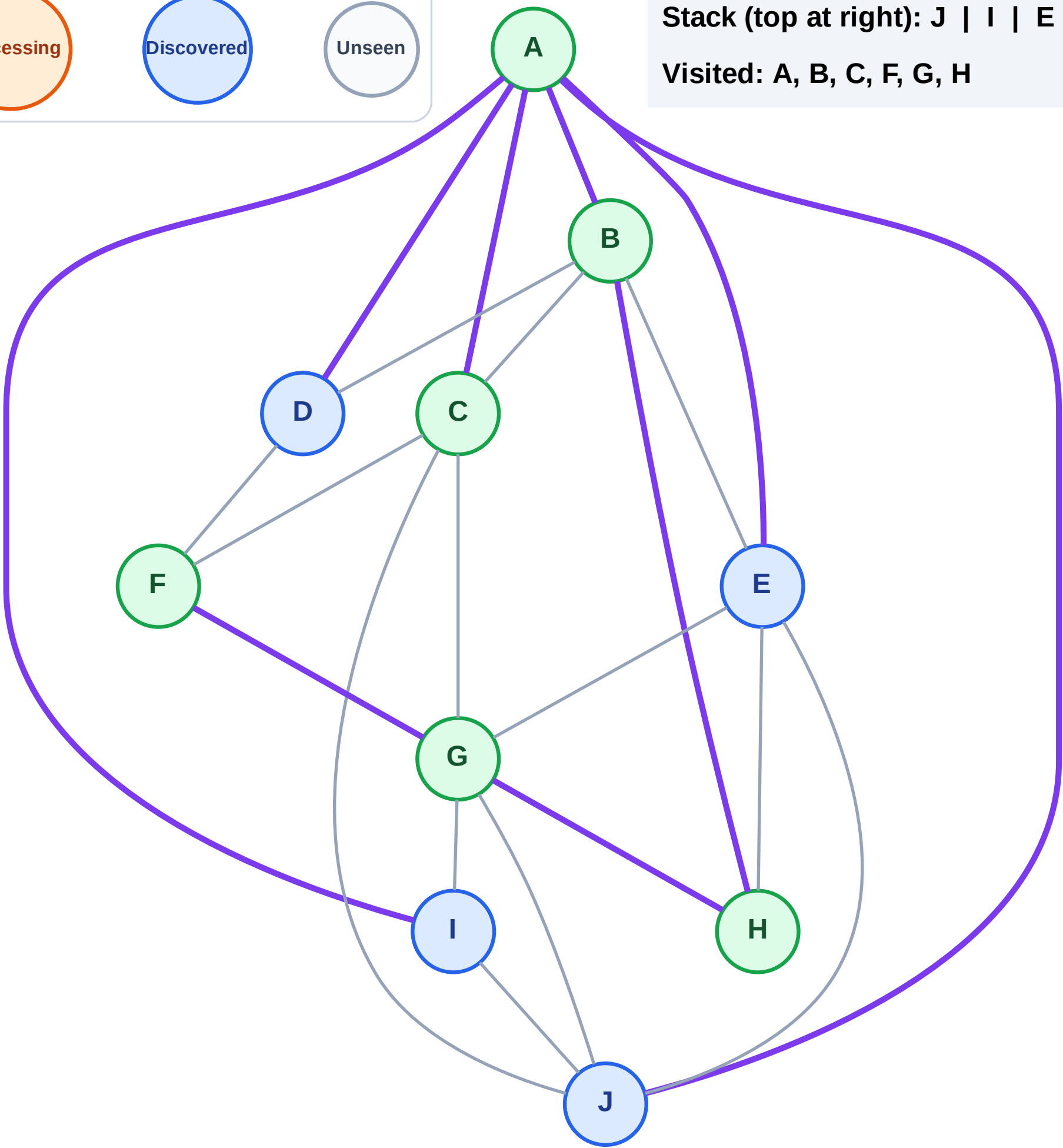
Step 13: No new neighbors from C

Legend



Stack (top at right): J | I | E | D

Visited: A, B, C, F, G, H



DFS Traversal

Step 14: Process node D

Legend

Complete

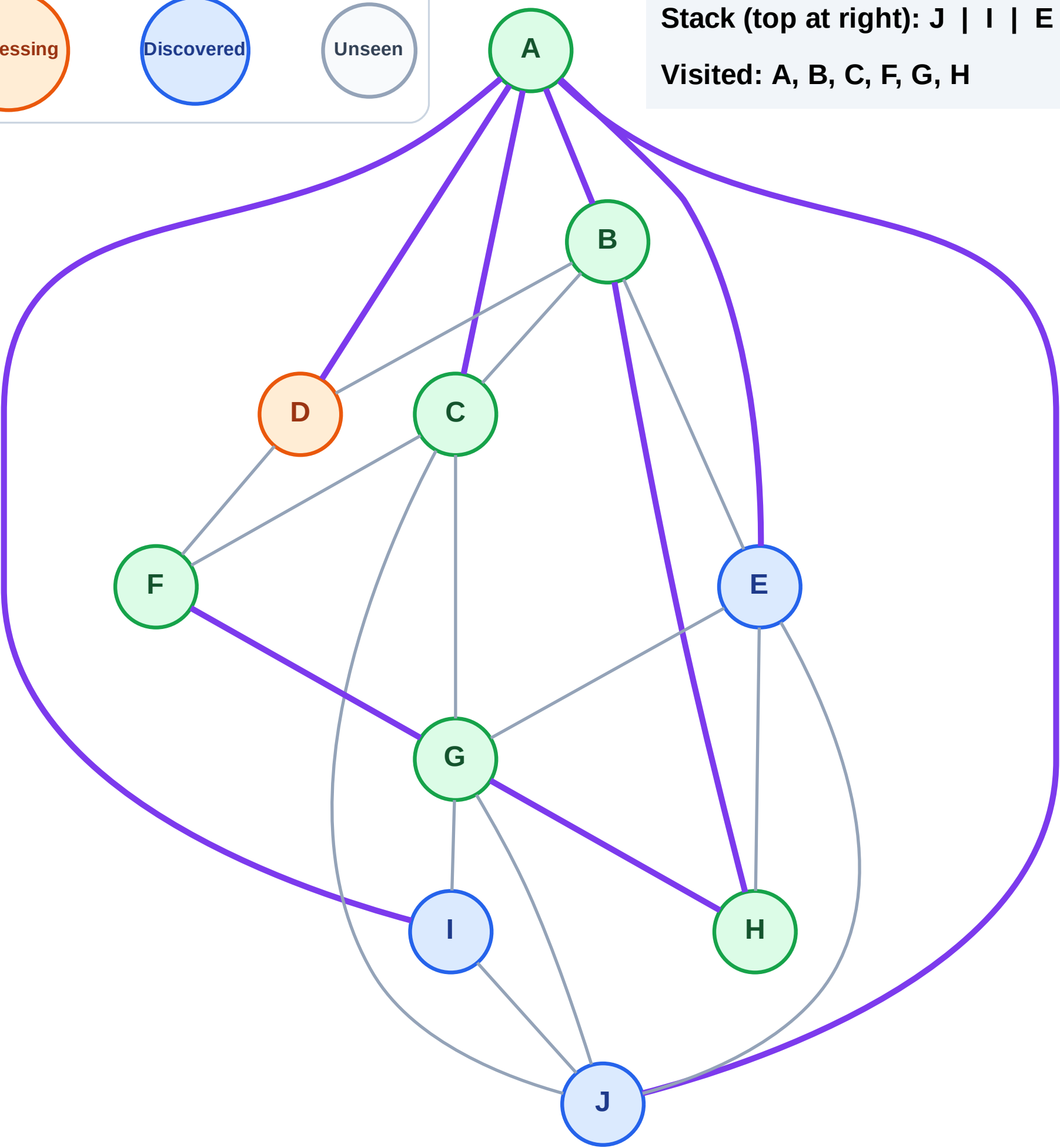
Processing

Discovered

Unseen

Stack (top at right): J | I | E

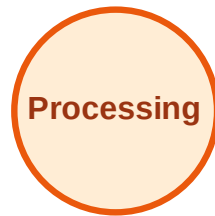
Visited: A, B, C, F, G, H



DFS Traversal

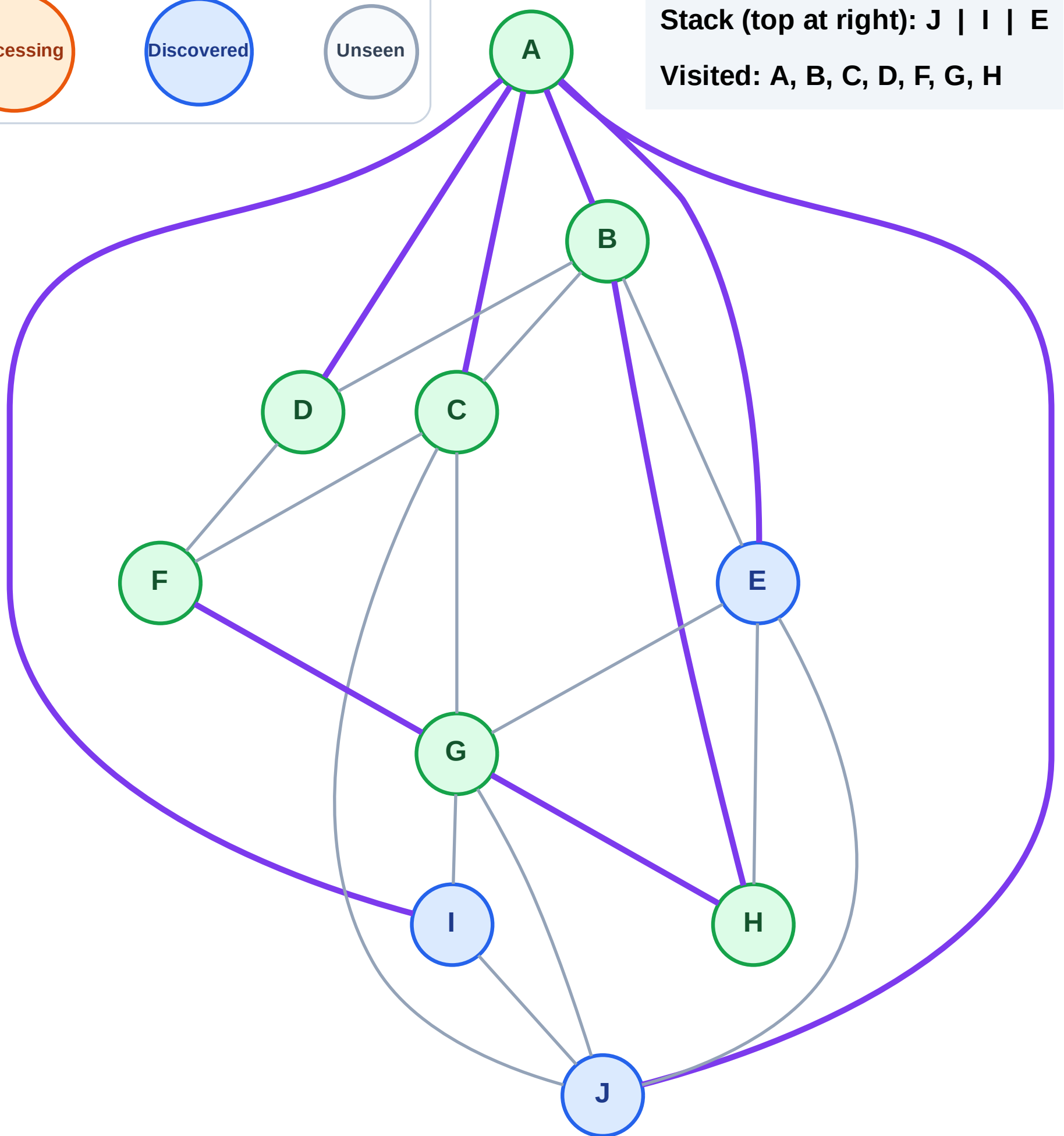
Step 15: No new neighbors from D

Legend



Stack (top at right): J | I | E

Visited: A, B, C, D, F, G, H



DFS Traversal

Step 16: Process node E

Legend

Complete

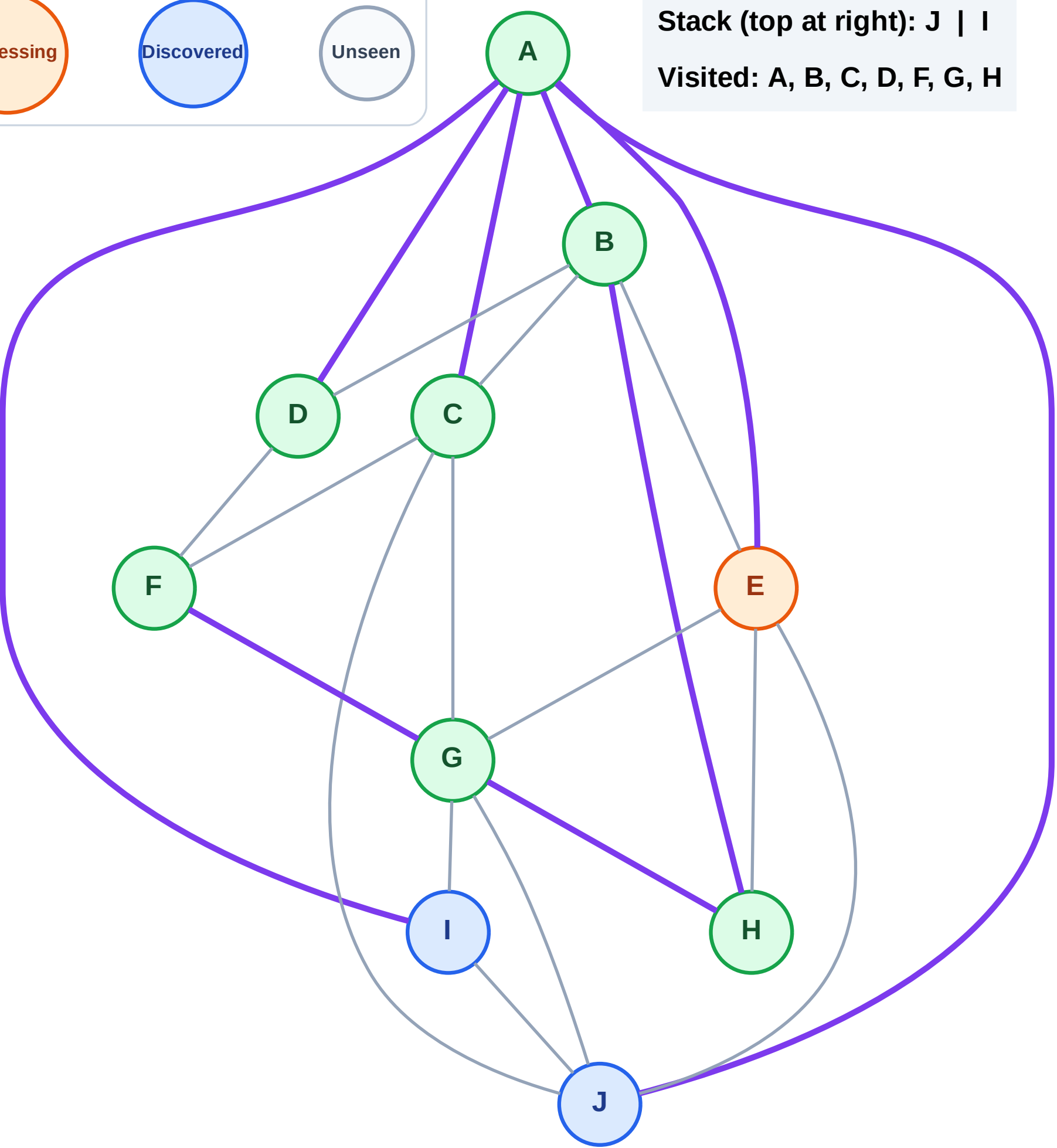
Processing

Discovered

Unseen

Stack (top at right): J | I

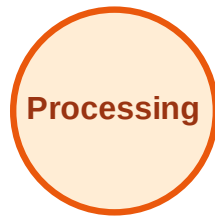
Visited: A, B, C, D, F, G, H



DFS Traversal

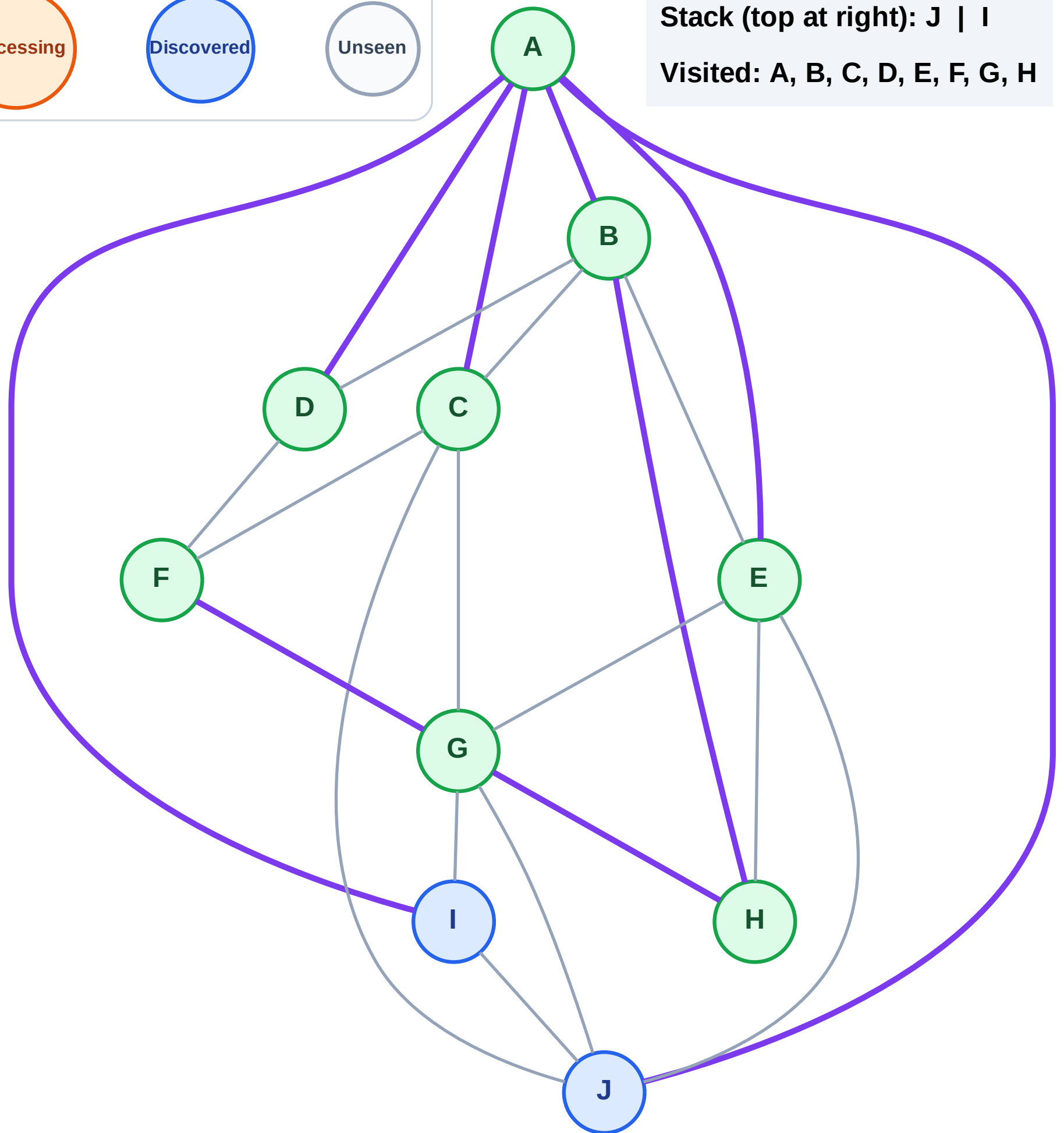
Step 17: No new neighbors from E

Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 18: Process node I

Legend

Complete

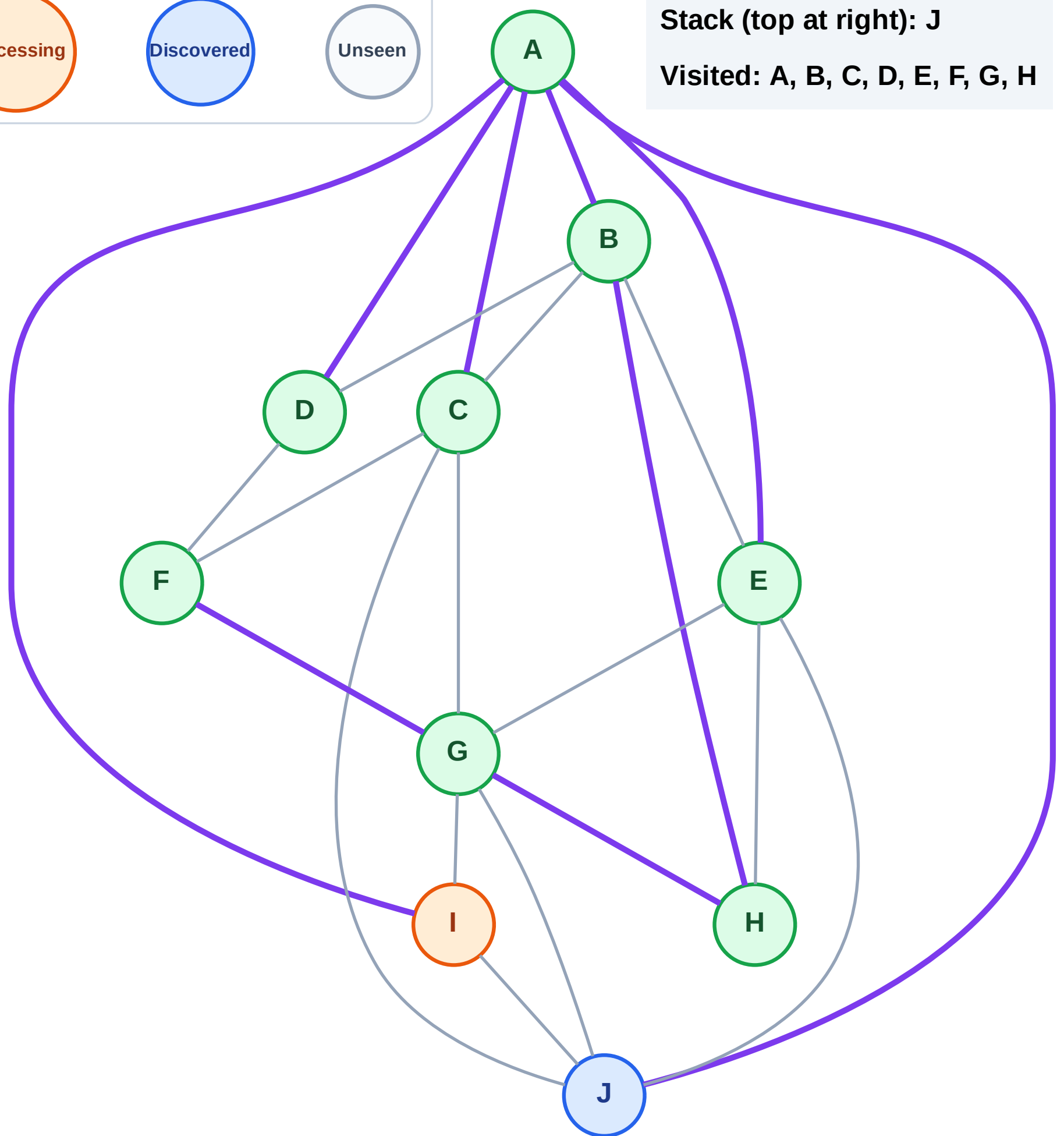
Processing

Discovered

Unseen

Stack (top at right): J

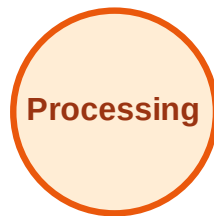
Visited: A, B, C, D, E, F, G, H



DFS Traversal

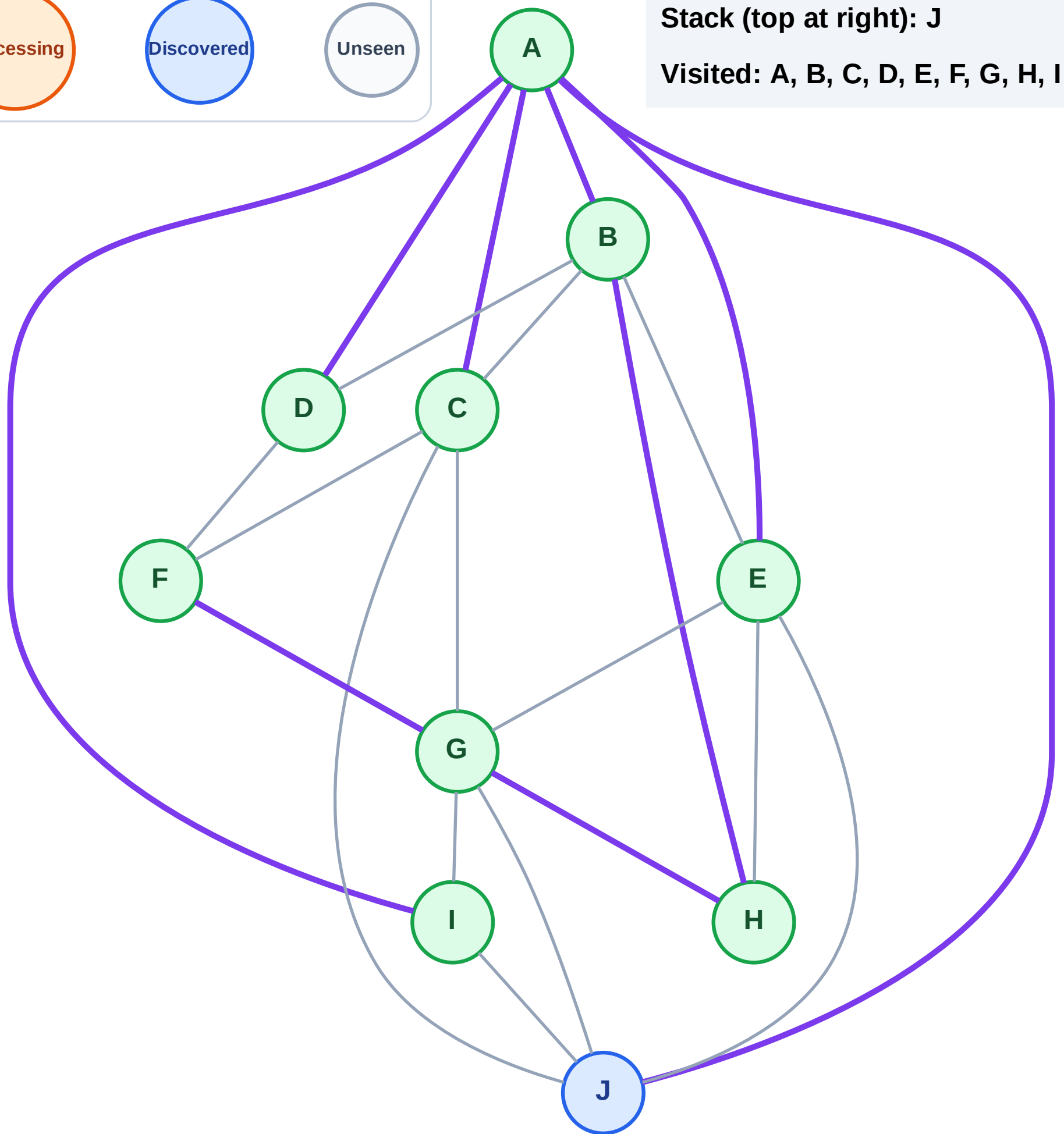
Step 19: No new neighbors from I

Legend



Stack (top at right): J

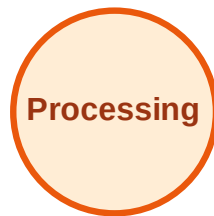
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

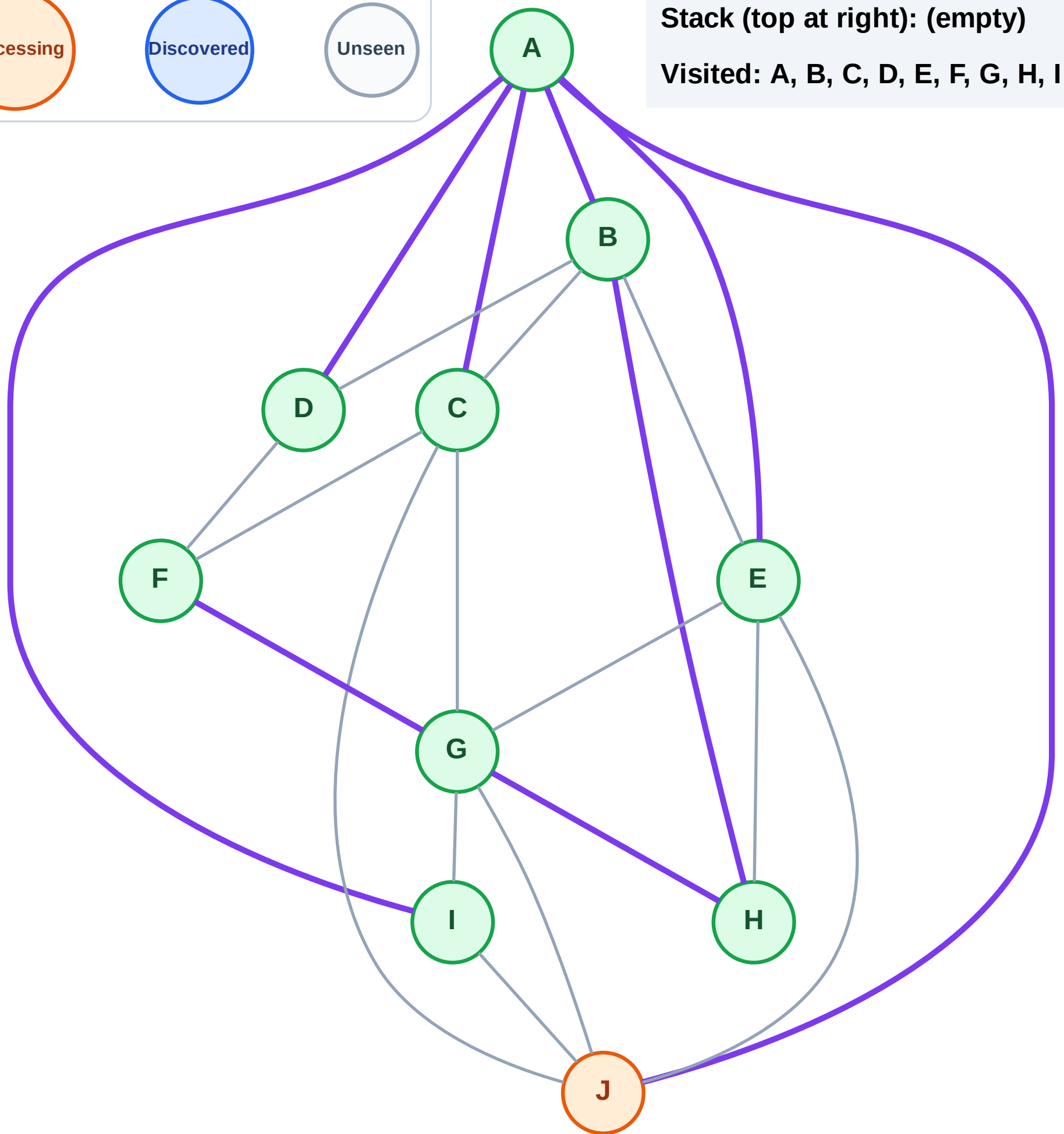
Step 20: Process node J

Legend



Stack (top at right): (empty)

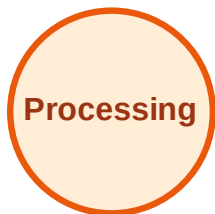
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

